

NATURAL MEASURES AND STATISTICAL PROPERTIES OF NON-STATISTICAL MAPS WITH MULTIPLE NEUTRAL FIXED POINTS

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Abstract. We show that a large class of infinite measure preserving dynamical systems that do not admit physical measures nevertheless exhibit strong statistical properties. In particular, we give sufficient conditions for existence of a distinguished natural measure ν such that the pushforwards of any absolutely continuous probability measure converge to ν . Moreover, we obtain a distributional limit law for empirical measures. Both of these are new results for intermittent maps with at least two neutral fixed points preserving an infinite σ -finite absolutely continuous measure.

We also extend existing results on the characterisation of the set of almost sure limit points for empirical measures. This result is new when there are at least three neutral fixed points.

1. Introduction

Let $f : X \rightarrow X$ be a dynamical system and let m be a reference probability measure with respect to which we wish to study the statistical behaviour of almost every point. A basic object encoding the long-term behaviour of an orbit is the sequence of *empirical measures*

$$(1.1) \quad e_n(x) := \frac{1}{n} \sum_{j=0}^{n-1} \delta_{f^j x} = \frac{1}{n} \sum_{j=0}^{n-1} f_*^j \delta_x.$$

Understanding the asymptotic behaviour of $e_n(x)$ for typical points is one of the central problems in ergodic theory.

If m is invariant under f , then Birkhoff's ergodic theorem guarantees that $e_n(x)$ converges for m -almost every x .¹ If, however, m is not invariant, then very different scenarios may occur. One common situation is that the system admits finitely many invariant probability measures which describe the statistical behaviour of m -almost every point. In this case, the long-term dynamics is governed by *physical measures*: an invariant probability measure ω_0 is called physical if there exists $E \subset X$ with $m(E) > 0$ such that $e_n(x) \rightarrow \omega_0$ for all $x \in E$. This framework encompasses a large class of dynamical systems studied in the literature (see [ABV00; AV18; Bow08; BR75; Dol00; Gan+21; Yan21]).

More recently, there has been significant interest in *nonstatistical maps*², namely systems for which $e_n(x)$ fails to converge for m -almost every x ; see [AG22; Bar22; BC25; BM08; CYZ20; KS17; Tal25] and references therein. These systems lie at the opposite extreme from those described above and, in particular, admit no physical measures. Their statistical description therefore requires new approaches.

For nonstatistical maps there are three directions that we explore in this paper:

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¹Here and throughout, X is assumed to be a topological space, and convergence of sequences of measures is with respect to the weak-* topology unless otherwise specified.

²Nonstatistical behaviour is also referred to as historical behaviour in the literature

- (A) Determination of the set of a.s. limit points of e_n .
- (B) Characterisation of the distributional limits of e_n .
- (C) Weak-* convergence of $\frac{1}{n} \sum_{j=0}^{n-1} f_*^j \lambda$ where $\lambda \ll m$.

We have not seen direction (B) addressed before. Also, we obtain to our knowledge the first results for direction (A) in situations where the set of a.s. limit points is neither a singleton nor an interval. With regard to direction (C), a probability measure ν is called *natural* [BB03; CE12; JT05; Mis05] if there exists an absolutely continuous probability measure λ such that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=0}^{n-1} f_*^j \lambda = \nu.$$

(By the dominated convergence theorem, physical measures are natural.)

In this article, we show that a large class of infinite measure preserving dynamical systems that are nonstatistical, and hence do not admit physical measures nevertheless exhibit strong statistical properties along the lines of directions (A)–(C) above.

Our results apply in particular to *Pomeau-Manneville maps*, which were introduced in the 1980s by [PM80] as models for intermittent turbulent or chaotic bursts interspersed with laminar behaviour. Subsequently, they have provided prototypical examples of nonuniformly hyperbolic dynamics in smooth ergodic theory and are regarded as a test-bed for new theoretical approaches in the area. The most studied examples are the *Liverani-Saussol-Vaienti maps* [LSV99], which form a special case of the intermittent maps analysed in detail by Thaler [Tha80; Tha83].

In this paper, we focus on the maps considered by Thaler [Tha80] as well as more recent examples of Coates *et al.* [CLM23]. Specifically, we study maps that preserve an infinite σ -finite absolutely continuous measure and have several “equally sticky” neutral fixed points.

For the sake of clarity, we first introduce our results for Thaler maps with two neutral fixed points [Tha02] before discussing more general situations. Let $f : [0, 1] \rightarrow [0, 1]$ be an interval map with the following properties for some $\alpha \in (0, 1]$, $b_1, b_2 > 0$, $c \in (0, 1)$:

- (1) $0, 1$ are neutral fixed points: $f'(0) = f'(1) = 1$;
- (2) $f|_{(0,c)}$ and $f|_{(c,1)}$ are C^2 diffeomorphisms onto $(0, 1)$ admitting C^2 extensions onto $[0, 1]$;
- (3) $f'(x) > 1$ for all $x \in (0, 1)$ and f is convex (resp. concave) on a neighbourhood of 0 (resp. 1);
- (4) $fx - x \sim b_1 x^{1+1/\alpha}$ as $x \rightarrow 0$ and $fx - x \sim b_2 (1-x)^{1+1/\alpha}$ as $x \rightarrow 1$.

By [Tha83, Eq. (1.5)], f is conservative and ergodic with a unique (up to scaling) invariant σ -finite absolutely continuous measure μ . Moreover, by [Tha83, p. 71], $\mu([0, 1]) = \infty$. However, it was shown in [ATZ05, Theorem 2 and discussion thereafter] that for almost every $x \in [0, 1]$, the sequence of empirical measures $e_n(x)$ diverges for almost every $x \in [0, 1]$ and hence f is non-statistical. Moreover, the set of weak-* limit points is almost surely equal to the set

$$\mathcal{S} := \{\nu_p := p\delta_0 + (1-p)\delta_1 : p \in [0, 1]\}$$

of all convex combinations of the Dirac masses at the neutral fixed points.

In the absence of physical measures, one may ask whether convergence holds when the point mass δ_x in (1.1) is replaced by other types of probability measure λ and whether there is a distinguished limit measure $\nu_{\bar{p}} \in \mathcal{S}$.

Our first result (corresponding to direction C above) shows that there exists a unique natural measure $\nu_{\bar{p}} \in \mathcal{S}$. Moreover, $\nu_{\bar{p}}$ attracts all absolutely continuous probability measures λ and it is not necessary to take Cesàro averages. (See [Kel04] for similar results in the simpler symmetric setting with $b_1 = b_2$, where $\bar{p} = \frac{1}{2}$. See also [Zwe02] for results on nonexistence of natural measures when f is badly behaved near the neutral fixed points.)

Theorem 1.1 (Existence and uniqueness of natural measures). *There exists a $\bar{p} \in (0, 1)$, with corresponding measure $\nu_{\bar{p}} \in \mathcal{S}$, such that*

$$\lim_{n \rightarrow \infty} f_*^n \lambda = \nu_{\bar{p}} \quad \text{for every absolutely continuous probability measure } \lambda.$$

Remark 1.2. A formula for $\bar{p}$ is given in Section 4.1, namely $\bar{p} = c_1/(c_1 + c_2)$ where c_1, c_2 are as in (4.5).

An almost immediate consequence of Theorem 1.1 is the following decay of correlations type result.

Corollary 1.3. *Suppose that $\varphi : [0, 1] \rightarrow \mathbb{R}$ is bounded, measurable, and continuous at 0 and 1. Then*

$$\lim_{n \rightarrow \infty} \int \psi \cdot \varphi \circ f^n d\text{Leb} = \int \psi d\text{Leb} \int \varphi d\nu_{\bar{p}} \quad \text{for all } \psi \in L^1([0, 1], \text{Leb}).$$

Equivalently, $\lim_{n \rightarrow \infty} \int \psi \cdot \varphi \circ f^n d\mu = \int \psi d\mu \int \varphi d\nu_{\bar{p}}$ for all $\psi \in L^1([0, 1], \mu)$.

We now return to consideration of the sequence e_n of empirical measures. Since e_n fails to converge pointwise, it is natural to consider alternative modes of convergence such as distributional convergence (direction B above).

To describe the distributional convergence of e_n , we recall the classical arcsine law for occupation times of [Lév39] as adapted by [Lam58] and then by [Tha02] to the current context. Define the sequence of occupation times

$$S_n := \sum_{j=0}^{n-1} 1_B \circ f^j,$$

where $B \subset [0, 1)$ is a closed interval containing 0. Also, we define the $[0, 1]$ -valued random variable $Z_{\alpha, p}$ for $\alpha \in (0, 1)$, $p \in [0, 1]$, as follows: When $\alpha = 1$, we set $Z_{1, p} \equiv p$. When $\alpha \in (0, 1)$, let $Z_{\alpha, p}$ be the random variable with continuous density

$$(1.2) \quad \frac{\hat{p} \sin \alpha \pi}{\pi} \frac{t^\alpha (1-t)^{\alpha-1} + t^{\alpha-1} (1-t)^\alpha}{\hat{p}^2 t^{2\alpha} + 2\hat{p} t^\alpha (1-t)^\alpha \cos \alpha \pi + (1-t)^{2\alpha}},$$

where $\hat{p} = p^{-1} - 1$. Thaler [Tha02, Theorem, p. 1291] proved a distributional limit law for occupation times:

$$\frac{1}{n} S_n \text{ converges strongly in distribution to } Z_{\alpha, \bar{p}}.$$

When $\alpha \in (0, 1)$, this means that $\lambda(\frac{1}{n} S_n \in I) \rightarrow \mathbb{P}(Z_{\alpha, \bar{p}} \in I)$ for all absolutely continuous probability measures λ on $[0, 1]$ and all open intervals $I \subset [0, 1]$.³ (When $\alpha = 1$, we exclude intervals I with an endpoint at $\bar{p}$.) Equivalently, $\frac{1}{n} S_n * \lambda \rightarrow \omega_0$ where ω_0 is the distribution $\omega_0(I) = \mathbb{P}(Z_{\alpha, \bar{p}} \in I)$ of $Z_{\alpha, \bar{p}}$.

To each $[0, 1]$ -valued random variable Z , we can associate the $\mathcal{S}$ -valued random variable $\nu_Z = Z\delta_0 + (1-Z)\delta_1$.

³We use standard probabilistic conventions, so on the left-hand side $\lambda(\frac{1}{n} S_n \in I) = \lambda(x \in [0, 1] : \frac{1}{n} S_n(x) \in I)$ and on the right-hand side $\mathbb{P}(Z_{\alpha, \bar{p}} \in I)$ is the integral of the density (1.2) over I .

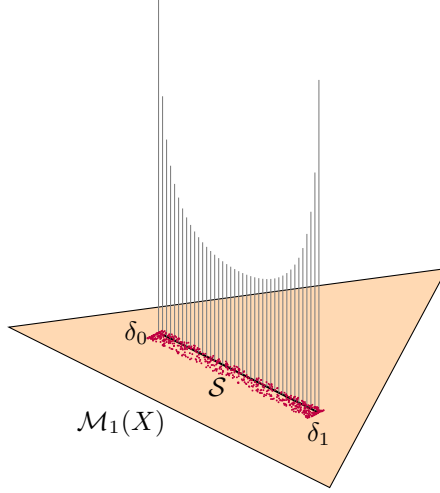


Figure 1. A schematic picture for the distributional limit law for empirical measures. The small dots around the set S are empirical measures $e_n(x)$ of an ensemble of points in the phase space chosen randomly (w.r.t. an absolutely continuous probability measure) for some fixed large n . As n increases, each individual dot oscillates between δ_0 and δ_1 ([ATZ05]), but in such a way as to make together an asymptotic distribution on S (Theorem 1.4).

Theorem 1.4 (Distributional limit law for empirical measures).

$$e_n \text{ converges strongly in distribution to } \nu_{Z_{\alpha, \bar{p}}}.$$

Theorem 1.4 describes how the $e_n(x)$ are asymptotically distributed on S , see Figure 1.

Let $\mathcal{M}_1(M)$ denote the space of probability measures on a measure space M . The sequence $e_n : [0, 1] \rightarrow \mathcal{M}_1([0, 1])$ induces the pushforward sequence $e_{n*} : \mathcal{M}_1([0, 1]) \rightarrow \mathcal{M}_1(\mathcal{M}_1([0, 1]))$. Since $(e_{n*}\lambda)(\mathcal{E}) = \lambda(e_n \in \mathcal{E})$, we can view Theorem 1.4 equivalently as a statement about the limit of $e_{n*}\lambda$ for all absolutely continuous measures λ :

Corollary 1.5. *Let $\omega \in \mathcal{M}_1(\mathcal{M}_1([0, 1]))$ be the distribution of $\nu_{Z_{\alpha, \bar{p}}}$. Then $\lim_{n \rightarrow \infty} e_{n*}\lambda = \omega$ for every absolutely continuous probability measure $\lambda \in \mathcal{M}_1([0, 1])$.*

Since $e_{n*}\lambda$ is a probability measure on the convex set $\mathcal{M}_1([0, 1])$, we can consider its expectation $\mathbb{E}(e_{n*}\lambda) = \int_{\mathcal{M}_1([0, 1])} \omega_0 de_{n*}\lambda \in \mathcal{M}_1([0, 1])$, which can be interpreted as a probability measure on $[0, 1]$ satisfying $\mathbb{E}(e_{n*}\lambda)(E) = \int \omega_0(E) de_{n*}\lambda(\omega_0)$.

We have the following consequence of Theorem B:

Corollary 1.6. *Let λ be an absolutely continuous probability measure on $[0, 1]$. Then*

$$(a) \frac{1}{n} \sum_{j=0}^{n-1} f_*^j \lambda = \int e_n d\lambda = \mathbb{E}(e_{n*}\lambda) \text{ for all } n \geq 1.$$

$$(b) \text{ The common limit of the sequences of measures in (a) is given by } \lim_{n \rightarrow \infty} \mathbb{E}(e_{n*}\lambda) = \nu_{\bar{p}}.$$

Beyond two neutral fixed points. Our results generalise to any finite number of neutral fixed points $\xi_1, \dots, \xi_d$. The proof of Theorem 1.1 requires no modifications. The treatment of the arcsine law in [Tha02; TZ06] assumes $d = 2$ but this restriction is removed by Sera & Yano [SY19] using a different method. We use [SY19] to generalise Theorem 1.4. The generalisations of Theorem 1.1 and 1.4 are stated in Section 2 as Theorems C and B respectively.

The final main result in this paper is Theorem A which shows that for general $d \geq 1$ and $\alpha \in (0, 1)$, the set of limit points of the sequence of empirical measures e_n is almost surely equal to the $(d - 1)$ -dimensional simplex

$$\mathcal{S} := \{p_1\delta_{\xi_1} + \dots + p_d\delta_{\xi_d} : p_1, \dots, p_d \geq 0, p_1 + \dots + p_d = 1\}.$$

Previous arguments in [ATZ05; CL24] exploited connectedness of the set of limit points and were effective for $d \leq 2$, $\alpha \in (0, 1]$. Our method uses instead the full support of the arcsine law in [SY19] combined with Theorem B. However, this full support property fails for $\alpha = 1$ which means that Theorem A remains open for $d \geq 3$, $\alpha = 1$.

The remainder of the paper is organised as follows. In Section 2, we present an abstract setup which allows for any finite number of neutral fixed points, and we state our main results Theorem A, B and C. Also, we state and prove Corollaries 2.8 and 2.9 which generalise Corollaries 1.6 and 1.3 respectively. Theorems A, B and C are proved in Section 3. In Section 4, we verify that our abstract setup includes the classes of intermittent maps in [CLM23; Tha80].

Notation. We use “big O” and $\lesssim$ notation interchangeably, writing $a_n = O(b_n)$ or $a_n \lesssim b_n$ if there are constants $C > 0$, $n_0 \geq 1$ such that $a_n \leq Cb_n$ for all $n \geq n_0$. As usual, $a_n = o(b_n)$ means that $a_n/b_n \rightarrow 0$ and $a_n \sim b_n$ means that $a_n/b_n \rightarrow 1$.

2. Abstract statement of results

2.1. Maps with Gibbs-Markov first return maps. In this section we recall some definitions and basic properties of maps with Gibbs-Markov first return maps needed to state our main assumptions.

Let X be a compact metric space with Borel probability measure m and let $f : X \rightarrow X$ be a nonsingular measurable map, so $m(f^{-1}E) = 0$ for every null set E . We suppose that f is ergodic, so for every measurable set E with $f^{-1}E = E$ we have $m(E) = 0$ or 1. Also, we suppose that f is conservative, which is equivalent to requiring that $f^n x \in E$ infinitely often for a.e. $x \in E$ for any positive measure set E . (A standard reference for this material is [Aar97, Ch 1].)

We say that the map f together with a countable partition $\mathcal{P}$ of X into positive measure subsets is a *Markov map* if the restriction of f to each partition element is a measurable bijection onto a union of partition elements. For $\theta \in (0, 1)$, we define the symbolic metric $d_\theta(x, y) := \theta^{s(x, y)}$ where $s(x, y) := \inf\{n \geq 0 : f^n x, f^n y \text{ lie in different elements of } \mathcal{P}\}$. It is assumed that $s(x, y) = \infty$ if and only if $x = y$ so that d_θ is a metric.

A Markov map f is said to be *topologically mixing* (w.r.t. the symbolic metric topology) if for every pair of open nonempty subsets $U, V \subset X$, there exists an $N \geq 1$ such that $f^n U \cap V \neq \emptyset$ for every $n \geq N$. Equivalently, for every $a, b \in \mathcal{P}$ there exists an $N \geq 1$ such that $f^n a \cap b \neq \emptyset$ for every $n \geq N$.

A standard approach in dynamical systems is *inducing*, whereby one studies the complex dynamics of f on X by focusing on the “nicer” return dynamics to a well-chosen inducing domain $Y \subset X$. We take $Y \subset X$ to be a union of partition elements and define the *first return time* $\tau : Y \rightarrow \mathbb{Z}^+$ and *first return map* $F : Y \rightarrow Y$ by setting

$$\tau(y) := \inf\{n \geq 1 : f^n y \in Y\}, \quad F(y) := f^{\tau(y)} y.$$

(Conservativity ensures that τ is finite a.e. and hence that F is well-defined a.e.)

Let $\mathcal{P}_Y$ be the partition of Y consisting of sets $(\bigcap_{j=0}^{n-1} f^{-j} a_i) \cap f^{-n} Y$ where $a_0, \dots, a_{n-1} \in \mathcal{P}$ with $a_0 \subset Y$ and $a_1, \dots, a_{n-1} \subset X \setminus Y$. Then F is Markov with partition $\mathcal{P}_Y$.

For the map F , we define the symbolic metric $d_\theta = \theta^s$ w.r.t. the partition $\mathcal{P}_Y$. The map F is *Gibbs-Markov* if

- **finite images:** $\text{Card}\{Fa : a \in \mathcal{P}_Y\} < \infty$;
- **bounded distortion:** there exists $\theta \in (0, 1)$ and $C > 0$ such that the function $\log \frac{dm}{dm \circ F}$ is d_θ -Lipschitz on elements of $\mathcal{P}_Y$.

Denote by $\mathcal{B}_\theta(Y)$ the space of real-valued functions which are Lipschitz continuous on Y with respect to the metric d_θ . We recall the following classical result about the existence of invariant measures for maps with Gibbs-Markov first return maps.

Lemma 2.1. *Suppose that $f : X \rightarrow X$ is a conservative ergodic Markov map with Gibbs-Markov first return map $F = f^\tau : Y \rightarrow Y$. Then f preserves a unique (up to scaling) absolutely continuous σ -finite measure μ , and μ is equivalent to m .*

Moreover, the density $h = d\mu/dm$ is nonzero and bounded on Y and $h|_Y, (h|_Y)^{-1} \in \mathcal{B}_\theta(Y)$. In particular, $\mu(Y) < \infty$.

Furthermore, $\mu(X) < \infty$ if and only if $\tau \in L^1(Y, m)$.

Proof. Standard references include [ADU93] and [Aar97, Chapter 4]. □

2.2. Assumptions and notations. Let X be a compact metric space with Borel probability measure m . Throughout, we suppose that $f : X \rightarrow X$ is a conservative, exact, Markov map with Gibbs-Markov first return map $F = f^\tau : Y \rightarrow Y$, and moreover that f and F are topologically mixing. Let μ be the f -invariant absolutely continuous σ -finite measure in Lemma 2.1. We assume that $\mu(X) = \infty$. Let $d \geq 1$ be a positive integer.

H1: There exist fixed points $\xi_1, \dots, \xi_d \in X$ such that

$$\mu(X \setminus \{B_\varepsilon(\xi_1) \cup \dots \cup B_\varepsilon(\xi_d)\}) < \infty \quad \text{for all } \varepsilon > 0.$$

This assumption implies that neighbourhoods of the fixed points $\xi_1, \dots, \xi_d$ carry the infinite part of the mass.

The next assumption ensures that the inducing set Y *dynamically separates* $\xi_1, \dots, \xi_d$ and that the excursion times from Y to neighbourhoods of the fixed points have certain tail distributions.

H2: There exist constants $\alpha \in (0, 1]$, $c_1, \dots, c_d > 0$ and a measurable partition $X_1, \dots, X_d$ of $X \setminus Y$ such that

- (a) $\xi_k \in \text{Int } X_k$, for $k = 1, \dots, d$;
- (b) For $k \neq \ell$, orbits cannot pass from X_k to X_ℓ without first entering Y . Equivalently,

$$\{\tau = n\} = \bigcup_{k=1, \dots, d} \{\tau^{(k)} = n - 1\} \quad \text{for } 1 \leq n < \infty$$

where $\tau^{(k)}(y) := \text{Card}\{n \leq \tau(y) : f^n y \in X_k\}$, for $y \in Y$, $k = 1, \dots, d$;

- (c) $\mu(\tau^{(k)} > n) \sim c_k n^{-\alpha}$ as $n \rightarrow \infty$ for $k = 1, \dots, d$.

Note that

$$(2.1) \quad \mu(\tau > n) \sim c_\tau n^{-\alpha} \quad \text{where} \quad c_\tau := c_1 + \dots + c_d > 0.$$

Remark 2.2. We expect that the results in this paper remain valid when the constants c_k in H2(c) are replaced by $c_k \ell(n)$ where ℓ is *slowly varying*, i.e. $\lim_{x \rightarrow \infty} \ell(\lambda x)/\ell(x) = 1$ for all $\lambda \in \mathbb{R}$. However, the proof of Theorem C requires further calculations.

Our third assumption is somewhat technical to state but is very mild. Given $a_0, \dots, a_r \in \mathcal{P}$, we define the $(r+1)$ -cylinder $[a_0, \dots, a_r] = \bigcap_{j=0}^r f^{-j}a_j$. Let $\mathcal{P}_r$ denote the partition of X into $(r+1)$ -cylinders and let $\mathcal{P}_r^*$ consist of $(r+1)$ -cylinders $a = [a_0, \dots, a_r] \in \mathcal{P}_r$ where $a_0, \dots, a_{r-1} \not\subset Y$ and $a_r \subset Y$. Given $\rho \in L^1(X, m)$ and $r \geq 0$, define

$$(2.2) \quad Q_r^\rho : Y \rightarrow \mathbb{R}, \quad Q_r^\rho := \sum_{a \in \mathcal{P}_r^*} 1_{f^r a} \cdot \frac{\rho}{Jf^r} \circ f^r|_a^{-1}$$

where $Jf^r := \frac{dm \circ f^r}{dm}$ denotes the Jacobian of f^r with respect to m .

H3: There exists $\theta \in (0, 1)$ and a dense subset $K(X) \subset L^1(X, m)$ such that $Q_r^\rho \in \mathcal{B}_\theta(Y)$ for all $\rho \in K(X)$ and $r \geq 0$.

Our final assumption is the following “smooth tail” estimate:

H4: If $\alpha \in (0, 1/2]$, then $\mu(\tau = n) = O(n^{-(1+\alpha)})$.

2.3. Almost sure behaviour of the empirical measures. We let $\mathcal{M}_1(X)$ denote the space of Borel probability measures on X . Define the map $e_n : X \rightarrow \mathcal{M}_1(X)$ which maps each point $x \in X$ to its n^{th} empirical measure $e_n(x)$ as given in (1.1).

Our first result concerns the almost sure behaviour of the empirical measures e_n . For $x \in X$, we let $\mathcal{L}(x) \subset \mathcal{M}_1(X)$ denote the set of weak-* limit points of $e_n(x)$. Let $\mathcal{S}_0$ be the simplex

$$\mathcal{S}_0 := \{p \in [0, 1]^d : p_1 + \dots + p_d = 1\}.$$

For $p \in \mathcal{S}_0$, let $\nu_p := p_1 \delta_{\xi_1} + \dots + p_d \delta_{\xi_d}$ be the corresponding convex combination of Dirac masses at the fixed points, and define

$$\mathcal{S} := \{\nu_p : p \in \mathcal{S}_0\},$$

to be the set of all such convex combinations. With this notation in place we can state our first result.

Theorem A. *Suppose that H1–H2 hold and that $\alpha \in (0, 1)$. Then*

$$\mathcal{L}(x) = \mathcal{S} \quad \text{for a.e. } x \in X.$$

In particular, when $d \geq 2$ there are no physical measures.

Remark 2.3. The inclusion $\mathcal{L}(x) \subset \mathcal{S}$ is elementary, see Lemma 3.2 below, so it is the reverse inclusion that is of interest. The case $d = 1$ is trivial: for all $\alpha \in (0, 1]$, we have $\mathcal{L}(x) = \delta_{\xi_1}$ a.e. and hence δ_{ξ_1} is a physical measure. The case $d = 2$ is also well-understood. Indeed, there are techniques [ATZ05; CL24] for showing that $\delta_{\xi_k} \in \mathcal{L}(x)$ a.e. for each k and Theorem A then follows for $d = 2$ by connectedness of $\mathcal{L}(x)$. This approach includes the case $\alpha = 1$.

For $d \geq 3$, Theorem A is completely new to the best of our knowledge. Our proof involves a different technique which works for all d but only in the range $\alpha \in (0, 1)$. We expect that Theorem A holds also for $d \geq 3$, $\alpha = 1$, but new ideas seem to be required.

2.4. Distributional convergence of the empirical measures. Our second result concerns strong distributional convergence of the sequence e_n of empirical measures.

Notation: Suppose that Z_n is a sequence of measurable functions on X taking values in some Borel space M and that Z is a random variable with distribution $\omega \in \mathcal{M}_1(M)$ also taking values in M (but not necessarily defined on X). Given $\lambda \in \mathcal{M}_1(X)$, we write $Z_n \rightarrow_\lambda Z$ if $\lim_{n \rightarrow \infty} \int Z_n \lambda = \int Z \omega$.

Definition 2.4. Let $\mathcal{M}_1^{\text{ac}}(X)$ denote the set of Borel probability measures on X which are absolutely continuous with respect to m . We say that Z_n *converges strongly in distribution* to Z , and write $Z_n \rightarrow_d Z$, if $Z_n \rightarrow_\lambda Z$ for all $\lambda \in \mathcal{M}_1^{\text{ac}}(X)$.

Fix ε so that the neighbourhoods $B_\varepsilon(\xi_k)$, $k = 1, \dots, d$ are disjoint. As was the case for the Thaler maps in Theorem 1.4, we first consider strong distributional convergence for occupation times $S_n = (S_n^1, \dots, S_n^d) : X \rightarrow [0, 1]^d$ defined by

$$(2.3) \quad S_n^k := \sum_{j=0}^{n-1} 1_{B_\varepsilon(\xi_k)} \circ f^j, \quad \text{for } k = 1, \dots, d, n \geq 1.$$

Following [SY19; Ser20], we consider a multiray generalisation of the classical arcsine law. For $\alpha \in (0, 1)$, $p \in \mathcal{S}_0$, let $\zeta_1, \dots, \zeta_d$ be independent $[0, \infty)$ -valued random variables defined on a common probability space with one-sided α -stable distribution characterised for $k = 1, \dots, d$ by

$$(2.4) \quad \mathbb{E} \exp(-t\zeta_k) = \exp(-t^\alpha p_k), \quad t > 0.$$

Define the $\mathcal{S}_0$ -valued random variable

$$(2.5) \quad Z_{\alpha,p} = (Z_{\alpha,p}^{(1)}, \dots, Z_{\alpha,p}^{(d)}) := \frac{1}{\zeta_1 + \dots + \zeta_d} (\zeta_1, \dots, \zeta_d).$$

When $\alpha = 1$, we define $Z_{1,p} \equiv p$.

Remark 2.5. In the special case that $d = 2$, one has that $Z_{\alpha,p}^{(2)} = 1 - Z_{\alpha,p}^{(1)}$ and for $\alpha \in (0, 1)$ the distribution of $Z_{\alpha,p}^{(1)}$ admits the continuous density in (1.2) (identifying p with p_1).

Remark 2.6. The first moment of the multiray arcsine law is given by $\mathbb{E} Z_{\alpha,p} = p$. This can be verified using the double Laplace formula [SY19, Proposition 2.6],

$$\int_0^\infty e^{-qt} \mathbb{E} \left(\exp \left\{ -t \sum_{k=1}^d \lambda_k \zeta_k \right\} \right) dt = \frac{\sum_{k=1}^d p_k (q + \lambda_k)^{\alpha-1}}{\sum_{k=1}^d p_k (q + \lambda_k)^\alpha}, \quad \lambda = (\lambda_1, \dots, \lambda_d) \in [0, \infty)^d, q > 0,$$

as derived in [Yan17, Proposition 3.6]. Differentiating w.r.t. λ_k and setting $\lambda = 0$ yields $\mathbb{E} Z_{\alpha,p}^{(k)} = p_k$.

We can now recall the arcsine law of [SY19; Ser20]. Recall from (2.1) that $c_\tau = \sum_{k=1}^d c_k$. Set

$$\bar{p} = (\bar{p}_1, \dots, \bar{p}_d) := c_\tau^{-1} (c_1, \dots, c_d) \in \mathcal{S}_0.$$

Theorem 2.7. *Suppose that H1–H2 hold. Then $\frac{1}{n} S_n \rightarrow_d Z_{\alpha,\bar{p}}$.*

Proof. It suffices to check that H1–H2 imply [Ser20, Assumptions 2.1, 2.2 and 2.6] as the result then follows from [Ser20, Corollary 4.2 and Theorem 3.3]. Let $F : Y \rightarrow Y$ be the Gibbs-Markov first return map in H2. Notice that H2(b) yields [Ser20, Assumption 2.1], and then H2(c) together with [Ser20, Lemma 2.4] gives [Ser20, Assumption 2.2]. Finally, as F is topologically mixing it is exponentially continued fraction mixing (see for example [Aar97, Section 4]) which immediately implies [Ser20, Assumption 2.6]. $\square$

To each $\mathcal{S}_0$ -valued random variable Z , we can associate the $\mathcal{S}$ -valued random variable $\nu_Z = Z_1 \delta_{\xi_1} + \dots + Z_d \delta_{\xi_d}$. Notice that $\mathbb{E} \nu_Z = \sum_{k=1}^d \mathbb{E} Z_k \delta_{\xi_k} = \sum_{k=1}^d (\mathbb{E} Z)_k \delta_{\xi_k} = \nu_{\mathbb{E} Z}$. In particular, $\mathbb{E} \nu_{Z_{\alpha,p}} = \nu_p$.

Theorem B. *Suppose that H1–H2 hold. Then $e_n \rightarrow_d \nu_{Z_{\alpha,\bar{p}}}$. Equivalently, $\lim_{n \rightarrow \infty} e_{n*} \lambda = \omega$ for all $\lambda \in \mathcal{M}_1^{\text{ac}}(X)$ where $\omega \in \mathcal{M}_1(\mathcal{S})$ is the distribution of $\nu_{Z_{\alpha,\bar{p}}}$.*

As in the introduction, we obtain the following consequence:

Corollary 2.8. *Let $\lambda \in \mathcal{M}_1^{\text{ac}}(X)$. Then*

(a) $\frac{1}{n} \sum_{j=0}^{n-1} f_*^j \lambda = \int e_n d\lambda = \mathbb{E}(e_n \lambda)$ for all $n \geq 1$.

(b) The common limit of the sequences of measures in (a) is given by $\lim_{n \rightarrow \infty} \mathbb{E}(e_n \lambda) = \nu_{\bar{p}}$.

Proof. Let $i : \mathcal{M}_1(X) \rightarrow \mathcal{M}_1(X)$ be the identity map. Then

$$\mathbb{E}(e_n \lambda) = \int_{\mathcal{M}_1(X)} \omega_0 d(e_n \lambda)(\omega_0) = \int_{\mathcal{M}_1(X)} i(\omega_0) d(e_n \lambda)(\omega_0) = \int_X i \circ e_n(x) d\lambda(x) = \int_X e_n d\lambda.$$

Next,

$$\left(\int_X \delta_{f^j x} d\lambda(x) \right)(E) = \int_X \delta_x(E) df_*^j \lambda(x) = \int_X 1_E df_*^j \lambda = (f_*^j \lambda)(E),$$

so $\int_X \delta_{f^j x} d\lambda(x) = f_*^j \lambda$. Hence

$$\int_X e_n d\lambda = \frac{1}{n} \sum_{j=0}^{n-1} \int_X \delta_{f^j x} d\lambda(x) = \frac{1}{n} \sum_{j=0}^{n-1} f_*^j \lambda.$$

(b) Since $\mathbb{E}$ acts continuously from $\mathcal{M}_1(\mathcal{M}_1(X))$ to $\mathcal{M}_1(X)$, it follows from Theorem B that $\lim_{n \rightarrow \infty} \mathbb{E}(e_n \lambda) = \mathbb{E}\nu_{Z_{\alpha, \bar{p}}} = \nu_{\bar{p}}$. $\square$

2.5. Existence and uniqueness of natural measures. Our third result concerns weak-* convergence of pushforwards $f_*^n \lambda$ of absolutely continuous probability measures $\lambda \in \mathcal{M}_1^{\text{ac}}(X)$.

Theorem C. *Suppose that H1–H4 hold with $\alpha \in (0, 1]$. Then*

$$\lim_{n \rightarrow \infty} f_*^n \lambda = \nu_{\bar{p}} \quad \text{for all } \lambda \in \mathcal{M}_1^{\text{ac}}(X).$$

As in the introduction, we obtain the following result on decay of correlations:

Corollary 2.9. *Suppose $\varphi : X \rightarrow \mathbb{R}$ is bounded, measurable, and continuous at each $\xi_1, \dots, \xi_d$. Then*

$$\lim_{n \rightarrow \infty} \int \psi \cdot \varphi \circ f^n dm = \int \psi dm \int \varphi d\nu_{\bar{p}} \quad \text{for all } \psi \in L^1(X, m).$$

Equivalently, $\lim_{n \rightarrow \infty} \int \psi \cdot \varphi \circ f^n d\mu = \int \psi d\mu \int \varphi d\nu_{\bar{p}}$ for all $\psi \in L^1(X, \mu)$.

Proof. We begin by proving that the first limit holds.

By Theorem C, $f_*^n \lambda \rightarrow \nu_{\bar{p}}$ weak-*, so by definition $\int \varphi df_*^n \lambda \rightarrow \int \varphi d\nu_{\bar{p}}$ for all continuous $\varphi : X \rightarrow \mathbb{R}$. By [Bil95, Theorem 25.7], $\int \varphi df_*^n \lambda \rightarrow \int \varphi d\nu_{\bar{p}}$ for all bounded measurable functions $\varphi : X \rightarrow \mathbb{R}$ that are continuous except on a set of $\nu_{\bar{p}}$ -measure zero, which is precisely the class of observables in the statement of the corollary.

Fix such a φ . Let $\psi \in L^1(X, m)$ with $\int \psi dm \neq 0$ and write $\psi = \rho \int \psi dm$ where $\rho \in L^1(X, m)$ and $\int \rho dm = 1$. Let $d\lambda = \rho dm$. Then

$$(\int \psi dm)^{-1} \int \psi \cdot \varphi \circ f^n dm = \int \rho \cdot \varphi \circ f^n dm = \int \varphi \circ f^n d\lambda = \int \varphi df_*^n \lambda \rightarrow \int \varphi d\nu_{\bar{p}}.$$

The case $\int \psi dm = 0$ is dealt with by approximating, concluding the proof of the first limit.

If $\psi \in L^1(X, \mu)$, then $\psi h \in L^1(X, m)$ and the second limit follows from the first. Similarly, the first limit follows from the second. $\square$

3. Proofs

In this section, we prove the three main theorems in this paper. It turns out to be convenient to prove them in the order B, A, C.

3.1. Ergodic theorem. In this subsection, we recall the ergodic theorem for infinite measure systems and derive a consequence for the systems studied in this paper.

Lemma 3.1. *Assume that [H1](#) holds. If $\varphi \in L^1(X, \mu)$, then $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=0}^{n-1} \varphi \circ f^j = 0$ almost everywhere.*

Proof. This is an easy consequence of the Hopf ratio ergodic theorem, see for example [[Aar97](#), Exercise 2.2.1]. $\square$

Lemma 3.2. *Assume that [H1](#) holds. Then $\mathcal{L}(x) \subset \mathcal{S}$ for almost every $x \in X$.*

Proof. We provide the details for completeness (cf. [[ATZ05](#), p. 963]).

For $\varepsilon > 0$, define $X^{(\varepsilon)} := X \setminus \{B_\varepsilon(\xi_1) \cup \dots \cup B_\varepsilon(\xi_d)\}$. By [H1](#), $\mu(X^{(\varepsilon)}) < \infty$ for all $\varepsilon > 0$. Choose $\psi : X \rightarrow [0, 1]$ continuous and supported in $X^{(\varepsilon/2)}$ such that $\psi|_{X^{(\varepsilon)}} \equiv 1$. In particular, $\int_X \psi d\mu < \infty$ so, by Lemma 3.1, there exists $X' \subset X$ with $\mu(X \setminus X') = 0$ such that $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=0}^{n-1} \psi(f^j x) = 0$ for all $x \in X'$.

Suppose that $x \in X'$ and $\omega_0 \in \mathcal{L}(x)$, and choose a subsequence n_i such that $e_{n_i}(x) \rightarrow \omega_0$. Then $\frac{1}{n_i} \sum_{j=0}^{n_i-1} \psi(f^j x) \rightarrow \int_X \psi d\omega_0$, yielding $\omega_0(X^{(\varepsilon)}) \leq \int_X \psi d\omega_0 = 0$. Hence, elements of $\mathcal{L}(x)$ are supported in $B_\varepsilon(x_1) \cup \dots \cup B_\varepsilon(x_d)$ for all $x \in X'$. Since $\varepsilon > 0$ can be chosen arbitrarily small, the result follows. $\square$

Corollary 3.3. *Assume that [H1](#) holds. Suppose that $e_{n_k} \rightarrow_m Z$ for some $\mathcal{M}_1(X)$ -valued random variable Z and some subsequence n_k . Then Z takes values in $\mathcal{S}$.*

Proof. Let $\psi : \mathcal{M}_1(X) \rightarrow \mathbb{R}$ be continuous and supported in $\mathcal{M}_1(X) \setminus \mathcal{S}$. By the dominated convergence theorem,

$$\int_X \psi(e_{n_k}(x)) dm(x) \rightarrow \mathbb{E} \psi(Z).$$

But $\psi(e_{n_k}) \rightarrow 0$ a.e. by Lemma 3.2. Applying the dominated convergence theorem once more, $\mathbb{E} \psi(Z) = 0$, and the result follows. $\square$

Remark 3.4. Corollary 3.3 shows that the only possible distributional limit points of the sequence e_n are random variables of the form ν_Z where Z is an $\mathcal{S}_0$ -valued random variable.

We recall also the following consequence of exactness.

Lemma 3.5. *Assume that [H1](#) holds. Let $\lambda \in \mathcal{M}_1^{\text{ac}}(X)$. Suppose that $f_*^{n_k} \lambda \rightarrow \omega_0$ as $k \rightarrow \infty$ for some $\omega_0 \in \mathcal{M}_1(X)$ and some subsequence n_k . Then $\omega_0 \in \mathcal{S}$.*

Proof. We show that $\omega_0(E) = 0$ for all $E \subset X$ with $\mu(E) < \infty$. The result then follows from [H1](#). Our argument follows [[BL21](#), Appendix A].

Write $\rho = d\lambda/dm$ and recall that $h = d\mu/dm$. Then

$$f_*^n \lambda(E) = \int_X 1_E \circ f^n d\lambda = \int_X 1_E \circ f^n \cdot \rho dm = \int_X 1_E \circ f^n \cdot \tilde{\rho} d\mu,$$

where $\tilde{\rho} = \rho h^{-1}$ satisfies $\int_X \tilde{\rho} d\mu = \int_X \rho dm = 1$. Let $A \subset X$ with $\mu(A) \in (0, \infty)$. Then $g := \tilde{\rho} - \mu(A)^{-1}1_A \in L^1(X, \mu)$ with $\int_X g d\mu = 0$. Hence $\lim_{n \rightarrow \infty} \int_X 1_E \circ f^n \cdot g d\mu = 0$ by Lin's Theorem [Lin71]. Writing

$$\int_X 1_E \circ f^n \cdot \tilde{\rho} d\mu = \int_X 1_E \circ f^n \cdot g d\mu + \int_X 1_E \circ f^n \cdot \mu(A)^{-1}1_A d\mu,$$

we deduce that

$$\limsup_{k \rightarrow \infty} f_*^{n_k} \lambda(E) \leq \limsup_{k \rightarrow \infty} \left| \int_X 1_E \circ f^{n_k} d\mu \right| \mu(A)^{-1} = \mu(E) \mu(A)^{-1}$$

The result follows since $\mu(E) < \infty$ and $\mu(A) \in (0, \infty)$ is arbitrarily large. $\square$

3.2. Proof of Theorem B. Fix $\varepsilon > 0$ such that $B_\varepsilon(\xi_k)$ are disjoint for $k = 1, \dots, d$ and define $S_n : X \rightarrow [0, 1]^d$ as in (2.3).

Lemma 3.6. *Let Z be a random variable with values in S_0 . Then $e_n \rightarrow_m \nu_Z$ if and only if $n^{-1}S_n \rightarrow_m Z$.*

Proof. Define

$$\pi : M_1(X) \rightarrow [0, 1]^d, \quad \pi(\omega_0) := (\omega_0(B_\varepsilon(\xi_1)), \dots, \omega_0(B_\varepsilon(\xi_d))).$$

Note that π restricts to the natural identification $\nu_Z \mapsto Z$ between $\mathcal{S}$ and S_0 . Also, π is continuous at elements in $\mathcal{S}$ and satisfies $\pi(e_n) = \frac{1}{n}S_n$.

In particular, if $e_{n_k} \rightarrow_m \nu_Z$ for some subsequence n_k , then it is an immediate consequence of the continuity of π at ν_Z and the continuous mapping theorem that $\frac{1}{n}S_{n_k} = \pi(e_{n_k}) \rightarrow_m \pi(\nu_Z) = Z$ completing the proof in one direction.

The converse follows by a standard probabilistic argument. Suppose that $\frac{1}{n}S_n \rightarrow_m Z$. Let $\widehat{Z}$ be a distributional limit point for e_n , so $e_{n_k} \rightarrow_m \widehat{Z}$ for some subsequence n_k . By Remark 3.4, $\widehat{Z} = \nu_A$ for some random variable A with values in S_0 . By what we just proved, $\frac{1}{n}S_{n_k} \rightarrow_m A$. Hence, $A = Z$, so $\widehat{Z} = \nu_Z$ is the unique distributional limit point for e_n . This means that $e_n \rightarrow_m \nu_Z$. $\square$

Proof of Theorem B. By Theorem 2.7, $n^{-1}S_n \rightarrow_m Z_{\alpha, \bar{p}}$. Hence by Lemma 3.6, $e_n \rightarrow_m \nu_{Z_{\alpha, \bar{p}}}$.

Let d_W denote the *Wasserstein distance* on $\mathcal{M}_1(X)$,

$$d_W(\omega_0, \omega'_0) := \sup_{\varphi \in \text{Lip}_1} \left| \int \varphi d\omega_0 - \int \varphi d\omega'_0 \right|,$$

where $\text{Lip}_1 = \{\varphi : X \rightarrow \mathbb{R} : \text{Lip } \varphi + \|\varphi\|_\infty \leq 1\}$ and $\text{Lip } \varphi$ denotes the smallest Lipschitz constant of $\varphi : X \rightarrow \mathbb{R}$. Recall that d_W induces the weak-* topology on $\mathcal{M}_1(X)$. Also, $d_W(e_n \circ f, e_n) \leq \frac{2}{n}$, so the functions $e_n : X \rightarrow \mathcal{M}_1(X)$ satisfy the “asymptotic invariance” condition $d_W(e_n \circ f, e_n) \rightarrow_m 0$. Hence, we may apply [Zwe07, Theorem 1] to deduce from $e_n \rightarrow_m \nu_{Z_{\alpha, \bar{p}}}$ that $e_n \rightarrow_d \nu_{Z_{\alpha, \bar{p}}}$. $\square$

3.3. Proof of Theorem A. We first show that the limiting random variable $\nu_{Z_{\alpha, \bar{p}}}$ in Theorem B has full support in $\mathcal{S}$ when $\alpha \in (0, 1)$.

Lemma 3.7. *Let $\alpha \in (0, 1)$, $p \in S_0$ with $p_i > 0$ for all i . Then $\mathbb{P}(\nu_{Z_{\alpha, p}} \in B_\varepsilon(\nu)) > 0$ for all $\varepsilon > 0$, $\nu \in \mathcal{S}$.*

Proof. Let $\zeta_1, \dots, \zeta_d$ be the independent, α -stable random variables that appear in the definition (2.5) of $Z_{\alpha,p}$. Recall [No120, Lemma 1.1, Proposition 3.2] that as the ζ_k are non-negative α -stable random variables with Laplace transforms given by (2.4), their distributions are fully supported and continuous on the half line $[0, \infty)$.

For every $q \in \mathcal{S}_0$, $\varepsilon > 0$ the set

$$U := \left\{ x \in [0, \infty)^d : \frac{x_k}{x_1 + \dots + x_d} \in (q_k - \varepsilon, q_k + \varepsilon), \text{ for all } k = 1, \dots, d \right\}$$

contains a nonempty open rectangle $\prod_{k=1}^d (a_k, b_k)$. By definition of $Z_{\alpha,p}$ and the independence of the ζ_k ,

$$\begin{aligned} \mathbb{P}(Z_{\alpha,p} \in B_\varepsilon(q)) &= \mathbb{P}((\zeta_1, \dots, \zeta_d) \in U) \\ &\geq \mathbb{P}(\zeta_k \in (a_k, b_k) : k = 1, \dots, d) = \prod_{k=1}^d \mathbb{P}(\zeta_k \in (a_k, b_k)) > 0. \end{aligned}$$

It follows that $\mathbb{P}(\nu_{Z_{\alpha,p}} \in B_\varepsilon(\nu_q)) > 0$. □

Proof of Theorem A. By Lemma 3.2, $\mathcal{L}(x) \subset \mathcal{S}$ for almost every x . To prove the converse, let $\nu \in \mathcal{S}$ and consider the function

$$\varphi(x) := \liminf_{n \rightarrow \infty} d(e_n(x), \nu),$$

where d is any metric metrising the weak-* topology. We will show that $\varphi = 0$ almost everywhere and so $\nu \in \mathcal{L}(x)$ for almost every x . The result then follows since $\nu \in \mathcal{S}$ is arbitrary.

Notice that φ is invariant for f and so, by ergodicity, must be almost everywhere equal to some constant $c \geq 0$. Suppose for contradiction that $c > 0$, and let $0 < \varepsilon < c$. It then follows that $m\{x : e_n(x) \in B_\varepsilon(\nu) \text{ for infinitely many } n\} = 0$.

By Theorem B, $e_n \rightarrow_m \nu_{Z_{\alpha,\bar{p}}}$ with $\bar{p}_i > 0$ for all i . By the Portmanteau lemma together with Lemma 3.7,

$$\liminf_{n \rightarrow \infty} m(e_n \in B_\varepsilon(\nu)) \geq \mathbb{P}(\nu_{Z_{\alpha,\bar{p}}} \in B_\varepsilon(\nu)) > 0.$$

Hence

$$\begin{aligned} m\{x : e_n(x) \in B_\varepsilon(\nu) \text{ for infinitely many } n\} &= m\left(\bigcap_{n=1} \bigcup_{\ell \geq n} \{e_\ell \in B_\varepsilon(\nu)\}\right) \\ &= \lim_{n \rightarrow \infty} m\left(\bigcup_{\ell \geq n} \{e_\ell \in B_\varepsilon(\nu)\}\right) \\ &\geq \liminf_{n \rightarrow \infty} m(e_n \in B_\varepsilon(\nu)) > 0, \end{aligned}$$

where we have used the fact that the sequence of sets $A_n = \bigcup_{\ell \geq n} \{e_\ell \in B_\varepsilon(\nu)\}$ is decreasing: $A_{n+1} \subset A_n$. This contradicts our assumption that $c > 0$ and so $\varphi(x) = 0$ almost everywhere. □

3.4. Proof of Theorem C.

Let f satisfy H1–H4.

We let $L : L^1(X, \mu) \rightarrow L^1(X, \mu)$ be the *transfer operator* for f defined by the relation

$$\int_X Lv \cdot w \, d\mu = \int_X v \cdot w \circ f \, d\mu, \quad \text{for all } v \in L^1(X, \mu), \, w \in L^\infty(X).$$

Define for $n \geq 0$,

$$T_n : L^1(Y, \mu|_Y) \rightarrow L^1(Y, \mu|_Y), \quad T_n v = 1_Y L^n(1_Y v).$$

We recall some results from [Goul1; MT12] which describe the asymptotic behaviour of T_n acting on the space $\mathcal{B}_\theta(Y)$ in Section 2.1.

Theorem 3.8. *Let $c_\tau \in (0, \infty)$ be as in (2.1). Then, for every $v \in \mathcal{B}_\theta(Y)$,*

$$\begin{aligned} c_\tau \log n T_n v &\rightarrow \int_Y v d\mu, & \alpha = 1 \\ c_\tau n^{1-\alpha} T_n v &\rightarrow \frac{1}{\pi} \sin \pi \alpha \int_Y v d\mu, & \alpha \in (0, 1) \end{aligned}$$

uniformly on Y as $n \rightarrow \infty$.

Proof. By assumption, $F = f^\tau : Y \rightarrow Y$ is a Gibbs-Markov map and $\mu(\tau > n) \sim c_\tau n^{-\alpha}$. Also, the underlying conservative ergodic map $f : X \rightarrow X$ is topologically mixing.

In the range $\alpha \in (\frac{1}{2}, 1]$ we can apply [MT12, Theorem 2.1, Proposition 11.4].

In the range $\alpha \in (0, \frac{1}{2}]$, we use [Gou11, Theorem 1.4]. The hypotheses in [Gou11] are stated slightly differently than in [MT12]. In the notation of these papers, the essential differences are as follows: (i) There is a stronger assumption on $\|R_n\|$ which holds since $\mu(\tau = n) = O(n^{-(\alpha+1)})$; (ii) There is the requirement that $R(1)$ has no eigenvalues on the unit circle besides 1 which holds since F is topologically mixing (see for example [AD01, Theorem 1.6]). $\square$

Define

$$Y_0 := Y; \quad \text{and} \quad Y_r := f^{-r}Y \setminus E_{r-1}, \quad \text{for } r \geq 1.$$

where $E_r := \bigcup_{i=0}^r f^{-i}Y$.

Recall that $h = \frac{d\mu}{dm}$ denotes the density of the invariant measure μ . Let $\lambda \in \mathcal{M}_1^{\text{ac}}(X)$ with density $\rho = \frac{d\lambda}{dm}$. Let $K(X)$ be the dense subset of $L^1(X, m)$ in H3.

Proposition 3.9. *If $\rho \in K(X)$, then $L^r(\rho h^{-1}1_{Y_r}) \in \mathcal{B}_\theta(Y)$ for all $r \geq 0$.*

Proof. Note first that $L^r(\rho h^{-1}1_{Y_r})$ is supported in Y by definition of Y_r . Let $M : L^1(X) \rightarrow L^1(X)$ denote the transfer operator for the reference measure m , so $L = h^{-1}Mh$. As usual, it follows from the definition $\int M^r v \cdot w dm = \int v \cdot w \circ f^r dm$ and change of variables that

$$M^r v = \sum_{a \in \mathcal{P}_r} 1_{f^r a} \cdot \frac{v}{J_r f} \circ f^r|_a^{-1}.$$

Hence, recalling (2.2),

$$\begin{aligned} L^r(\rho h^{-1}1_{Y_r}) &= h^{-1}M^r(\rho 1_{Y_r}) = h^{-1} \sum_{a \in \mathcal{P}_r} 1_{f^r a} \cdot 1_{Y_r} \circ f^r|_a^{-1} \cdot \frac{\rho}{J_r f} \circ f^r|_a^{-1} \\ &= h^{-1} \sum_{a \in \mathcal{P}_r^*} 1_{f^r a} \cdot \frac{\rho}{J_r f} \circ f^r|_a^{-1} = h^{-1}Q_r^\rho. \end{aligned}$$

By H3, $Q_r^\rho \in \mathcal{B}_\theta(Y)$, while $(h|_Y)^{-1} \in \mathcal{B}_\theta(Y)$ by Lemma 2.1. Hence $h^{-1}Q_r^\rho \in \mathcal{B}_\theta(Y)$. $\square$

For $r \geq 0$, define $\rho_r := \rho h^{-1}1_{E_r}$. Notice that

$$(3.1) \quad \rho_r = \sum_{j=0}^r \rho h^{-1}1_{Y_j}.$$

Corollary 3.10. *For all $\rho \in K(X)$ and $r \geq 0$,*

$$\begin{aligned} c_\tau \log n L^n \rho_r &\rightarrow \int_X \rho_r d\mu, & \alpha = 1 \\ c_\tau n^{1-\alpha} L^n \rho_r &\rightarrow \frac{1}{\pi} \sin \pi \alpha \int_X \rho_r d\mu, & \alpha \in (0, 1) \end{aligned}$$

uniformly on Y as $n \rightarrow \infty$.

Proof. We give the details in the case $\alpha = 1$. The case $\alpha \in (0, 1)$ is similar.

For $n \geq r \geq 0$, equation (3.1) gives

$$1_Y L^n \rho_r = \sum_{j=0}^r 1_Y L^n (\rho h^{-1} 1_{Y_j}) = \sum_{j=0}^r T_{n-j} L^j (\rho h^{-1} 1_{Y_j}).$$

Using again (3.1) and the fact that $\int_X \rho h^{-1} 1_{Y_j} d\mu = \int_Y L^j (\rho h^{-1} 1_{Y_j}) d\mu$, we obtain that on Y

$$\begin{aligned} c_\tau \log n L^n \rho_r - \int_X \rho_r d\mu &= \sum_{j=0}^r \frac{\log n}{\log(n-j)} \left\{ c_\tau \log(n-j) T_{n-j} L^j (\rho h^{-1} 1_{Y_j}) - \int_Y L^j (\rho h^{-1} 1_{Y_j}) d\mu \right\} \\ &\quad + \sum_{j=0}^r \left\{ \frac{\log n}{\log(n-j)} - 1 \right\} \int_X \rho h^{-1} 1_{Y_j} d\mu. \end{aligned}$$

The result follows by Theorem 3.8 and Proposition 3.9. $\square$

Now, let $X_1, \dots, X_d$ be the partition of $X \setminus Y$ in H2.

Lemma 3.11. *For all $\rho \in K(X)$, $n > r \geq 0$, $k = 1, \dots, d$,*

$$\int_X 1_{X_k} df_*^n(\lambda|_{E_r}) = \sum_{j=1}^n \int_{\{\tau^{(k)} \geq j\}} 1_Y L^{n-j} \rho_r d\mu.$$

Proof. Suppose that $x \in E_r$ and that $f^n x \in X_k$ for some $k = 1, \dots, d$. Then x must have made its last return to Y at some time $n - j$ for some $1 \leq j \leq n$. Moreover, by H2(b), $f^m x \in X_k$ for $n - j + 1 \leq m \leq n$. Hence,

$$\begin{aligned} \{x \in E_r : f^n x \in X_k\} &= \bigcup_{j=1}^n \{x \in E_r : f^{n-j} x \in Y \text{ and } f^m x \in X_k, n - j + 1 \leq m \leq n\} \\ &= \bigcup_{j=1}^n \{x \in E_r : f^{n-j} x \in Y \text{ and } \tau^{(k)}(f^{n-j} x) \geq j\}. \end{aligned}$$

As this is a disjoint union,

$$\begin{aligned} \int_X 1_{X_k} df_*^n(\lambda|_{E_r}) &= \int_X 1_{X_k} \circ f^n \cdot 1_{E_r} \cdot \rho \cdot h^{-1} d\mu \\ &= \sum_{j=1}^n \int_X (1_{\{\tau^{(k)} \geq j\}} 1_Y) \circ f^{n-j} \cdot 1_{E_r} \cdot \rho \cdot h^{-1} d\mu \\ &= \sum_{j=1}^n \int_{\{\tau^{(k)} > j-1\}} 1_Y L^{n-j} \rho_r d\mu. \end{aligned}$$

□

Proposition 3.12. *For all $\rho \in K(X)$, $r \geq 0$, $k = 1, \dots, d$,*

$$\lim_{n \rightarrow \infty} \int_X 1_{X_k} df_*^n(\lambda|_{E_r}) = \bar{p}_k \lambda \left(\bigcup_{i=0}^r f^{-i} Y \right).$$

Proof. Note that $\int \rho_r d\mu = \int_{E_r} \rho h^{-1} d\mu = \lambda(E_r)$.

First, we consider the case $\alpha \in (0, 1)$. Set

$$\varepsilon_n := \sup_Y \left| L^n \rho_r - \frac{d_\alpha \lambda(E_r)}{n^{1-\alpha}} \right|,$$

where $d_\alpha = \frac{\sin \pi \alpha}{\pi c_\tau}$. By Corollary 3.10, $\varepsilon_n = o(n^{\alpha-1})$, Hence, by Lemma 3.11,

$$\begin{aligned} \int 1_{X_k} df_*^n(\lambda|_{E_r}) &= \sum_{j=1}^n \int_{\{\tau^{(k)} \geq j\}} 1_Y L^{n-j} \rho_r d\mu \\ &= \sum_{j=1}^{n-1} \int_{\{\tau^{(k)} \geq j\}} 1_Y L^{n-j} \rho_r d\mu + O(\mu(\tau^{(k)} \geq n)) \\ (3.2) \quad &= d_\alpha \lambda(E_r) \sum_{j=1}^{n-1} \frac{\mu(\tau^{(k)} \geq j)}{(n-j)^{1-\alpha}} + O\left(\sum_{j=1}^{n-1} \frac{\varepsilon_{n-j} \mu(\tau^{(k)} \geq j)}{(n-j)^{1-\alpha}}\right) + O(n^{-\alpha}). \end{aligned}$$

As $\mu(\tau^{(k)} \geq j) \sim c_k j^{-\alpha}$ we can conclude from Lemma A.1 and Lemma A.2 that

$$(3.3) \quad \sum_{j=1}^{n-1} \frac{\mu(\tau^{(k)} \geq j)}{(n-j)^{1-\alpha}} \rightarrow c_k \frac{\pi}{\sin \pi \alpha} \quad \text{and} \quad \sum_{j=1}^{n-1} \frac{\varepsilon_{n-j} \mu(\tau^{(k)} \geq j)}{(n-j)^{1-\alpha}} \rightarrow 0.$$

Combining (3.2) and (3.3) we obtain

$$\int_{X_k} df_*^n(\lambda|_{E_r}) = \frac{c_k}{c_\tau} \lambda(E_r) + o(1) = \bar{p}_k \lambda(E_r) + o(1),$$

concluding the result in the case that $\alpha \in (0, 1)$.

When $\alpha = 1$, we proceed in the same manner as before. Set

$$\varepsilon_n := \sup_Y \left| L^n \rho_r - \frac{\lambda(E_r)}{c_\tau \log n} \right|$$

and note by Corollary 3.10 that $\varepsilon_n = o(1/\log n)$ Hence, by Lemma 3.11,

$$\begin{aligned} \int 1_{X_k} df_*^n(\lambda|_{E_r}) &= \sum_{j=1}^n \int_{\{\tau^{(k)} \geq j\}} 1_Y L^{n-j} \rho_r d\mu \\ &= \sum_{j=1}^{n-2} \int_{\{\tau^{(k)} \geq j\}} 1_Y L^{n-j} \rho_r d\mu + O(\mu(\tau^{(k)} \geq n-1)) \\ (3.4) \quad &= \frac{\lambda(E_r)}{c_\tau} \sum_{j=1}^{n-2} \frac{\mu(\tau^{(k)} \geq j)}{\log(n-j)} + O\left(\sum_{j=1}^{n-2} \frac{\varepsilon_{n-j} \mu(\tau^{(k)} \geq j)}{\log(n-j)}\right) + O(n^{-1}). \end{aligned}$$

Using Lemma A.3 and Lemma A.4 we conclude that

$$(3.5) \quad \sum_{j=1}^{n-2} \frac{\mu(\tau^{(k)} \geq j)}{\log(n-j)} \rightarrow c_k \quad \text{and} \quad \sum_{j=1}^{n-2} \frac{\varepsilon_{n-j} \mu(\tau^{(k)} \geq j)}{\log(n-j)} \rightarrow 0.$$

Combining (3.4) and (3.5) we obtain $\int_{X_k} df_*^n(\lambda|_{E_r}) = \bar{p}_k \lambda(E_r) + o(1)$, concluding the proof. $\square$

Proof of Theorem C. Since $K(X)$ is dense in $L^1(X, m)$, the conclusion of Proposition 3.12 holds for general $\lambda \in \mathcal{M}_1^{\text{ac}}(X)$.

Fix $k \in \{1, \dots, d\}$. Then

$$\left| \int_X 1_{X_k} df_*^n(\lambda|_{E_r}) - \int_X 1_{X_k} df_*^n \lambda \right| = \left| \int_X 1_{X_k} \circ f^n (1_X - 1_{E_r}) d\lambda \right| \leq \int_X |1_X - 1_{E_r}| d\lambda$$

for all $r \geq 1$, so by the dominated convergence theorem, $\lim_{r \rightarrow \infty} \int_X 1_{X_k} df_*^n(\lambda|_{E_r}) = \int_X 1_{X_k} df_*^n \lambda$ uniformly in n . Also, $\lim_{r \rightarrow \infty} \lambda(E_r) = 1$ by the dominated convergence theorem. Hence, for any $\varepsilon > 0$, there exists $r \geq 1$ such that $\left| \int_X 1_{X_k} df_*^n(\lambda|_{E_r}) - \int_X 1_{X_k} df_*^n \lambda \right| < \varepsilon$ for all $n \geq 1$ and $\bar{p}_k |\lambda(E_r) - 1| < \varepsilon$. By the first statement of the proof, we obtain that $\limsup_{n \rightarrow \infty} \left| \int_X 1_{X_k} \circ f^n d\lambda - \bar{p}_k \right| < 2\varepsilon$. Since $\varepsilon > 0$ is arbitrary, it follows that $\lim_{n \rightarrow \infty} \int_X 1_{X_k} \circ f^n d\lambda = \bar{p}_k$. In other words, $\lim_{n \rightarrow \infty} f_*^n \lambda(X_k) = \bar{p}_k$.

Let $\omega_0 = \lim_{i \rightarrow \infty} f_*^{n_i} \lambda$ be a subsequential limit of $f_*^n \lambda$. Then $\omega_0(X_k) = \bar{p}_k$ for each k . Also, $\omega_0 = \nu_{\bar{p}}$ for some $\bar{p} \in \mathcal{S}_0$ by Lemma 3.5. Hence $\omega_0 = \nu_{\bar{p}}$.

We have shown that $\nu_{\bar{p}}$ is the unique subsequential limit of $f_*^n \lambda$. By compactness of $\mathcal{M}_1(X)$, it follows that $\lim_{n \rightarrow \infty} f_*^n \lambda = \nu_{\bar{p}}$. $\square$

4. Examples

In this section, we apply our main results to intermittent maps. Throughout, we write $|E| = \text{Leb}(E)$ for measurable subsets $E \subset [0, 1]$.

4.1. Thaler maps. We define $\mathcal{T}$ to be the class of interval maps $f : [0, 1] \rightarrow [0, 1]$ studied in [Tha80; Tha83] which satisfy the following:

T1: There exists $0 = \xi_1 < \xi_2 < \dots < \xi_d = 1$, $d \geq 2$, with $f\xi_k = \xi_k$ and $f'(\xi_k) = 1$ for $k = 1, \dots, d$;

T2: There exist subintervals $I_1, \dots, I_d$, $d \geq 2$, with $\xi_k \in \text{Int } I_k$, such that $\bigcup_{k=1}^d I_k = [0, 1]$ and such that the restriction $f|_{I_k}$ extends to a C^2 diffeomorphism $f_k : \bar{I}_k \rightarrow [0, 1]$;

T3: $f'(x) > 1$ for all $x \notin \{\xi_1, \dots, \xi_d\}$ and there exists an $\varepsilon > 0$ so that f is concave (resp. convex) on the interval $(\xi_k - \varepsilon, \xi_k) \cap I_k$ (resp. $(\xi_k, \xi_k + \varepsilon) \cap I_k$);

T4: There exist $\alpha \in (0, 1]$ and $b_1, \dots, b_d > 0$ such that for every $k = 1, \dots, d$,

$$|fx - x| \sim b_k |x - \xi_k|^{1+1/\alpha} \quad \text{as } x \rightarrow \xi_k.$$

Remark 4.1. When $d = 2$, conditions T1–T4 reduce to conditions 1–4 given in the Introduction.

Thaler maps $f \in \mathcal{T}$ can be shown to lie in the abstract setting of Section 2 and hence Theorems A, B and C apply to these examples. The verification of the hypotheses in Section 2 is essentially contained in [Ser20, Proof of Theorem 4.6] and the references therein. For completeness we recall the main steps of this argument here. Different inducing schemes are needed in the cases $d = 2$ and $d \geq 3$, and the case $d = 2$ will be treated in the more general setting considered in Section 4.2, so we focus here on the case $d \geq 3$.⁴

By [Tha80; Tha83], $f \in \mathcal{T}$ is conservative and exact with a unique (up to scaling) invariant absolutely continuous measure μ ; moreover $\mu([0, 1]) = \infty$. Since the branches $f_k : \bar{I}_k \rightarrow [0, 1]$ are onto, it is immediate (see Section 2.1) that f is a topologically mixing Markov map.

⁴There are some typos in [Ser20, Equation (4.7)] where x_i should be x_j and f_j should be f_i .

Next, we describe the inducing scheme. Set

$$X_k := I_k \cap f^{-1}I_k, \quad k = 1, \dots, d; \quad \text{and} \quad Y := [0, 1] \setminus (X_1 \cup \dots \cup X_d).$$

Let $Y_k := I_k \setminus X_k$. Then $Y = Y_1 \dot{\cup} \dots \dot{\cup} Y_d$. Let $g_k := f_k^{-1} : [0, 1] \rightarrow \bar{I}_k$ denote the inverse branch of f on $\bar{I}_k$. Set $X_{k,n} := g_k^n Y_k$ for $n \geq 1, k = 1, \dots, d$. Notice that $\{X_{k,n}, n \geq 1\}$ is a partition of X_k and that the maps $f : X_{k,n+1} \rightarrow X_{k,n}, f : X_{k,1} \rightarrow Y_k$ are bijections. For $j \neq k$, set

$$Y_{j,k,1} := g_j Y_k \quad \text{and} \quad Y_{j,k,n} := g_j X_{k,n-1}, \quad n \geq 2.$$

Then $\{Y_{j,k,n} : j \neq k, n \geq 1\}$ is a partition of Y_j for each j and hence

$$\mathcal{P}_Y := \{Y_{j,k,n} : j \neq k, n \geq 1\}$$

is a partition of Y .

Let $F = f^\tau : Y \rightarrow Y$ be the first return map to Y . Then $\tau = n$ on $\bigcup_{j \neq k} Y_{j,k,n}$ and $F = f^n : Y_{j,k,n} \rightarrow Y_k$ is a bijection for all $j \neq k$ and all n .

Proposition 4.2. *The first return map $F : Y \rightarrow Y$ is a topologically mixing Gibbs-Markov map with respect to the partition $\mathcal{P}_Y$. In addition, [H1](#), [H2\(a,b\)](#) and [H3](#) hold.*

Proof. By construction, F is Markov with respect to the partition $\mathcal{P}_Y$. As $FY_{j,k,n} = Y_k$ it is clear that F has finite images. Moreover, as $FY_k = \bigcup_{\ell \neq k} Y_\ell$ we see that $F^3 Y_{j,k,n} = Y$ and so F is topologically mixing.

For $0 \leq m \leq n-1$ and $x, y \in [0, 1]$,

$$(4.1) \quad \log \frac{(f^{n-m})'(f^m x)}{(f^{n-m})'(f^m y)} = \sum_{i=m}^{n-1} \log \frac{f'(f^i x)}{f'(f^i y)} = \sum_{i=m}^{n-1} \frac{|f''(z_i)|}{f'(z_i)} |f^i x - f^i y| \leq |f''|_\infty \sum_{i=m}^{n-1} |f^i x - f^i y|,$$

where $z_i \in [f^i x, f^i y]$ is chosen by the mean value theorem. In particular, for $x, y \in Y_{j,k,n}$,

$$\log \frac{(f^{n-m})'(f^m x)}{(f^{n-m})'(f^m y)} \leq |f''|_\infty \left(|Y_{j,k,n}| + \sum_{i=1}^{\infty} |X_{k,i}| \right) \leq |f''|_\infty.$$

Hence for $i = 0, \dots, n-1$,

$$(4.2) \quad \frac{|f^i x - f^i y|}{|X_{k,n-i}|} \leq e^{2|f''|_\infty} \frac{|f^{n-i} f^i x - f^{n-i} f^i y|}{|f^{n-i} X_{k,n-i}|} = e^{2|f''|_\infty} \frac{|Fx - Fy|}{|Y_k|},$$

Inserting (4.2) into (4.1) with $m = 0$, we obtain that

$$(4.3) \quad \log \left| \frac{F'x}{F'y} \right| \leq C |Fx - Fy|$$

where $C = |f''|_\infty e^{2|f''|_\infty} \max_k |Y_k|^{-1}$.

Let $\lambda = \inf_Y f' > 1$ and set $\theta = \lambda^{-1}$. If $s(x, y) = n$, then $1 \geq |F^n x - F^n y| \geq \lambda^{n-1} |Fx - Fy|$. Combining this inequality with (4.3), we obtain $\log \left| \frac{F'x}{F'y} \right| \leq C\theta^{n-1} = C\theta^{-1} d_\theta(x, y)$, which concludes the proof that F is Gibbs-Markov.

We denote the density of μ by h . By [Tha83, Lemma 4], h is bounded on compact subsets of $[0, 1] \setminus \{\xi_1, \dots, \xi_d\}$. Hence, [H1](#) is satisfied. Also, it is immediate from the definitions that [H2\(a,b\)](#) hold.

Finally, we verify [H3](#) taking $K(X) = C^1(X)$. For this, we use very rough estimates (sharp estimates not depending on r are available but not needed). Let $\rho \in C^1(X)$ and note that

$$Q_r^\rho(y) = \sum \frac{\rho(x)}{(f^r)'(x)} \quad \text{for } y \in Y,$$

where the sum is over those $x \in X$ with $f^r x = y$ such that $x, fx, \dots, f^{r-1}x \notin Y$. There are at most d such preimages x , one in each X_k , so $|Q_r^\rho|_\infty \leq d|\rho|_\infty$. Next, given $y, y' \in Y$ we denote the corresponding preimages $x, x' \in X_k$. Then

$$(4.4) \quad |Q_r^\rho(y) - Q_r^\rho(y')| \leq \sum |\rho(x) - \rho(x')| + \sum |\rho|_\infty \left| \frac{1}{(f^r)'(x)} - \frac{1}{(f^r)'(x')} \right|.$$

Certainly,

$$\begin{aligned} |(f^r)'(x) - (f^r)'(x')| &= \left| \prod_{j=0}^{r-1} f'(f^j x) - \prod_{j=0}^{r-1} f'(f^j x') \right| \leq |f'|_\infty^{r-1} \sum_{j=0}^{r-1} |f'(f^j x) - f'(f^j x')| \\ &\leq |f'|_\infty^{r-1} |f''|_\infty \sum_{j=0}^{r-1} |f^j x - f^j x'| \leq r |f'|_\infty^{r-1} |f''|_\infty |f^r x - f^r x'| \leq r |f'|_\infty^{r-1} |f''|_\infty |y - y'|. \end{aligned}$$

It follows that

$$\left| \frac{1}{(f^r)'(x)} - \frac{1}{(f^r)'(x')} \right| \leq |(f^r)'(x) - (f^r)'(x')| \leq r |f'|_\infty^{r-1} |f''|_\infty |y - y'|.$$

Also, $|\rho(x) - \rho(x')| \leq |\rho'|_\infty |x - x'| \leq |\rho'|_\infty |y - y'|$. Hence,

$$|Q_r^\rho(y) - Q_r^\rho(y')| \leq d(|\rho'|_\infty + r|\rho|_\infty |f'|_\infty^{r-1} |f''|_\infty) |y - y'|.$$

As shown above, $|y - y'| \lesssim d_\theta(y, y')$, so $|Q_r^\rho(y) - Q_r^\rho(y')| \lesssim d_\theta(y, y')$, yielding $Q_r^\rho \in \mathcal{B}_\theta(Y)$ as required. $\square$

Let $e_k = 1$ for $k = 1, d$ and $e_k = 2$ for $2 \leq k \leq d-1$. Define

$$c_k := e_k b_k^{-\alpha} \alpha^\alpha \sum_{j=1, \dots, d, j \neq k} h(g_j \xi_k) g'_j(\xi_k).$$

In particular, for $d = 2$, we have

$$(4.5) \quad c_1 = b_1^{-\alpha} \alpha^\alpha h(g_2 0) g'_2(0), \quad c_2 = b_2^{-\alpha} \alpha^\alpha h(g_1 1) g'_1(1).$$

The remaining ingredients, namely [H2\(c\)](#) and [H4](#), follow from the next result.

Lemma 4.3. $\mu(\tau^{(k)} = n) \sim \alpha c_k n^{-(1+\alpha)}$ as $n \rightarrow \infty$ for $k = 1, \dots, d$.

Proof. Note that $\{\tau^{(k)} = n\} = \bigcup_{j=1, \dots, d, j \neq k} Y_{j,k,n-1}$.

Fix $j, k \in \{1, \dots, d\}$, $j \neq k$. For $k \neq 1, d$, the sets $X_{k,n}$ have two connected components $X_{k,n}^\pm$ which lie to the left or right of ξ_k and both accumulate at ξ_k as $n \rightarrow \infty$. Accordingly, define $Y_{j,k,n}^\pm := g_j X_{k,n-1}^\pm$. It suffices to show that $\mu(Y_{j,k,n}^\pm)$ for $k \neq 1, d$ and $\mu(Y_{j,k,n})$ for $k = 1, d$ have the asymptotic

$$b_k^{-\alpha} \alpha^{1+\alpha} h(g_j \xi_k) g'_j(\xi_k) n^{-(1+\alpha)} \quad \text{as } n \rightarrow \infty.$$

We give the details for $\mu(Y_{j,k,n}^+)$, $k \neq 1, d$, the other cases being similar.

By [\[Tha83, Lemma 4\]](#), h is continuous on $[0, 1] \setminus \{\xi_1, \dots, \xi_d\}$ (in the standard topology). Hence,

$$(4.6) \quad \mu(Y_{j,k,n}^+) - h(g_j \xi_k) |Y_{j,k,n}^+| = \int_{Y_{j,k,n}^+} (h - h(g_j \xi_k)) d\text{Leb} \leq |Y_{j,k,n}^+| \sup_{Y_{j,k,n}^+} |h - h(g_j \xi_k)|$$

It follows that $\mu(Y_{j,k,n}^+) \sim h(g_j \xi_k) |Y_{j,k,n}^+|$. By the mean value theorem, there exists $z_n \in X_{k,n}^+$ so that $|Y_{j,k,n}^+| = g'_j(z_n) |X_{k,n-1}^+| \sim g'_j(\xi_k) |X_{k,n-1}^+|$. Combining these last two estimates,

$$(4.7) \quad \mu(Y_{j,k,n}^+) \sim h(g_j \xi_k) g'_j(\xi_k) |X_{k,n-1}^+|.$$

It remains to estimate $|X_{k,n}^+|$. We recall the following standard calculation.

Sublemma 4.4 (see for example [Aar97, Lemma 4.8.6]). *Suppose that $T : [0, C] \rightarrow [0, \infty)$ is such that $Tx > x$ for each $x \in (0, C]$ and $Tx \sim x + bx^{1+p}$ as $x \rightarrow 0$. Let $z_n = Tz_{n+1}$. Then*

$$z_n \sim (pbn)^{-1/p} \quad \text{and} \quad z_n - z_{n+1} \sim b^{-1/p}(pn)^{-(1+1/p)} \quad \text{as } n \rightarrow \infty$$

Proof. It follows from the assumptions that the sequence z_n is strictly decreasing with $z_n \rightarrow 0$. We then calculate

$$z_n^{-p} \sim z_{n+1}^{-p}(1 + bz_{n+1}^p)^{-p} = z_{n+1}^{-p}(1 - pbz_{n+1}^p + o(z_{n+1}^{2p})) = z_{n+1}^{-p} - pb + o(1)$$

as $n \rightarrow \infty$. Summing over n , this yields $z_n^{-p} = pbn + o(n)$ which gives the estimate for z_n . Finally, $z_n - z_{n+1} = Tz_{n+1} - z_{n+1} \sim bz_{n+1}^{1+p} \sim b^{-1/p}(pn)^{-(1+1/p)}$. $\square$

Recalling T4, the estimate for $|X_{k,n}^+|$ reduces after a change of coordinates to the situation in Sublemma 4.4 with $p = 1/\alpha$ and $b = b_k$. Hence $|X_{k,n}^+| \sim b_k^{-\alpha}(n/\alpha)^{-(1+\alpha)}$. Combining this with (4.7), we obtain

$$\mu(Y_{j,k,n}^+) \sim h(g_j \xi_k) g_j'(\xi_k) b_k^{-\alpha} (\alpha/n)^{1+\alpha}$$

as required. $\square$

4.2. Intermittent maps with critical points and/or singularities. We now consider a class of intermittent interval maps with two branches that possibly admit a critical point and/or a singularity at the discontinuity. This class of maps include the maps in $\mathcal{T}$ with $d = 2$ and the maps described in [CLM23]. We define $\mathcal{F}$ to be set of maps $f : [-1, 1] \rightarrow [-1, 1]$ which satisfy the following conditions.

F0: There exists a $c \in (-1, 1)$ such that the restrictions $f_- := f : (-1, c) \rightarrow (-1, 1)$ and $f_+ : (c, 1) \rightarrow (-1, 1)$ are C^2 orientation preserving diffeomorphisms with no fixed points.

Remark 4.5. For notational simplicity we will assume that $c = 0$. Notice also that we do not assume that the $f_{\pm}$ extend to C^2 functions on the closure of $I_{\pm}$.

F1: There exist $\ell_+, \ell_- > 0$ and $k_+, k_- > 0$ such that

$$(4.8) \quad f(x) = \begin{cases} x + b_-(1+x)^{1+\ell_-} + o((1+x)^{1+\ell_-}), & \text{for } x \in U_{-1} \\ 1 - a_-|x|^{k_-}, & \text{for } x \in U_{0-} \\ -1 + a_+|x|^{k_+}, & \text{for } x \in U_{0+} \\ x - b_+(1-x)^{1+\ell_+} + o((1-x)^{1+\ell_+}), & \text{for } x \in U_{+1} \end{cases}$$

whenever $k_+, k_- \neq 1$ for some $a_{\pm}, b_{\pm} > 0$, and some neighbourhoods U_{-1}, U_{0-} of $-1, 0$ in $[-1, 0]$ and some neighbourhoods U_{+1}, U_{0+} of $1, 0$ in $[0, 1]$. If $k_+ = 1$ and/or $k_- = 1$, then we replace the corresponding lines in (4.8) with the assumption that $f'(0-) = a_- > 1$ and/or $f'(0+) = a_+ > 1$ respectively.

Remark 4.6. It is convenient to assume that $fU_{0\pm} \subset U_{\mp 1}$. Notice that this assumption poses no restriction on F1 as $U_{0\pm}$ can be taken to be arbitrarily small.

Remark 4.7. This definition is more general than the one in [CLM23] as F1 stipulates only an asymptotic behaviour near the fixed points. However, it does not include the maps in [MS22] due to the restriction mentioned in Remark 2.2. We expect that our results hold also for the maps in [MS22].

Suppose that f satisfies **F0** and let $\gamma_{\pm} \in I_{\pm}$ be the two points of period 2 for f . Define the intervals

$$(4.9) \quad Y := [\gamma_-, \gamma_+], \quad X_{\pm,n} := f_{\pm}^{-n}Y, \quad \text{and} \quad Y_{\pm,n+1} := f_{\pm}^{-1}X_{\mp,n}.$$

By definition, for $n \geq 1$, $f : Y_{\pm,n+1} \rightarrow X_{\mp,n}$ and $f : X_{\pm,n} \rightarrow X_{\pm,n-1}$ and $f : X_{\pm,1} \rightarrow Y$ are bijections. Moreover, from **F0**, we know that $X_{\pm,n}$ are consecutive intervals, accumulating at ± 1 . So, the $\{X_{\pm,n} : n \geq 1\}$ partition $X \setminus Y$ and, as $f(Y) = X \setminus Y$, the $\{Y_{\pm,n} : n \geq 2\}$ partition Y . Thus, $\mathcal{P} := \{Y_{\pm,j}, X_{\pm,n} : n \geq 1, j \geq 2\}$ forms a Markov partition for f . Our final condition will ensure that our maps have good expansion and distortion properties.

F2: f is convex (resp. concave) on U_{-1} (resp. U_1) and moreover

- (1) If f is not C^2 on $\overline{U}_{-1}$ (resp. $\overline{U}_{+1}$), then $f''(x) \lesssim (1+x)^{\ell_- - 1}$ (resp. $|f''(x)| \lesssim (1-x)^{\ell_+ - 1}$).
- (2) If $k_{\pm} \neq 1$, then there exists a $\lambda > 1$ such that $(f^n)'(x) > \lambda$ for each $x \in Y_{\pm,n}$ and each $Y_{\pm,n} \not\subset U_{0\pm}$.

Remark 4.8. Notice that **F2.1** is only assumed when the map is not C^2 at the fixed points and that **F2.2** is only an assumption about the map outside of the neighbourhoods $U_{0\pm}, U_{\pm 1}$ and is trivially satisfied if $f'(x) > 1$ for each $x \notin \{-1, 1\}$.

We let $\alpha_+ := 1/\ell_+ k_+$, $\alpha_- := 1/\ell_- k_+$ and define

$$c_1 := h(0)a_-^{-1/k_-}(\ell_+ b_+)^{-1/\alpha_-} \alpha_-^2, \quad \text{and} \quad c_2 := h(0)a_+^{-1/k_+}(\ell_- b_-)^{-1/\alpha_+} \alpha_+^2.$$

We will assume that $\alpha_+ = \alpha_- = \alpha \in (0, 1)$.

Theorem 4.9. *Suppose that $f \in \mathcal{F}$ with $\alpha \in (0, 1)$. Then **H1-H4** hold.*

Throughout this section we fix

$$X_- := [0, \gamma_-), \quad X_+ := (\gamma_+, 0],$$

and let $\tau : Y \rightarrow \mathbb{N}$ denote the first return time to Y and $\tau^{(\pm)}(x) := \text{Card}\{n \leq \tau(x) : f^n(x) \in X_{\pm}\}$. Let $F : Y \rightarrow Y$ be the first return map. By construction, we obtain the following lemma.

Lemma 4.10. $F|_{Y_{\pm,n}} = f^n$ and the interval Y dynamically separates X_- , X_+ .

In the remainder of this section we will show that f satisfies assumptions **H1-H3** and thus conclude the proof of Theorem 4.9. Notice that $X_{\pm,n}$ and $Y_{\pm,n}$ here play the same roles as $\Delta_n^{\pm}$ and $\delta_n^{\pm}$ respectively in **[CLM23]**. The next Lemma shows that the asymptotic behaviour of the sizes of these partition elements remains exactly the same as in **[CLM23]**.

Lemma 4.11.

$$|X_{\pm,n}| \sim b_{\pm}^{-1/\ell_{\pm}}(\ell_{\pm} n)^{-1-1/\ell_{\pm}} \quad \text{and} \quad \text{Leb}(\tau^{(\pm)} = n) = |Y_{\pm,n}| \sim a_{\pm}^{-1/k_{\pm}}(\ell_{\mp} b_{\mp})^{-\alpha_{\pm}} \alpha_{\pm} n^{-1-\alpha_{\pm}}.$$

Proof. We will only explicitly prove the estimates for $Y_{+,n}$ and $X_{-,n}$ as the other estimates follow in the same way. Let $\gamma_n := f_-^{-n}\gamma$ so that $X_{-,n} = [\gamma_n, \gamma_{n-1}]$. As in the previous section we can apply Sublemma 4.4 to the sequence $z_n := 1 + \gamma_n$ with $b = b_-$ and $p = \ell_-$ to obtain

$$(4.10) \quad z_n \sim (b_- \ell_- n)^{-1/\ell_-}, \quad \text{and} \quad |X_{\pm,n}| \sim b_{\pm}^{-1/\ell_{\pm}}(\ell_{\pm} n)^{-1-1/\ell_{\pm}}.$$

Now, notice that as $y \downarrow 0$ we have by definition $f_+^{-1}y = \left(\frac{y+1}{a_+}\right)^{1/k_+}$. It follows that $f_+^{-1}\gamma_n = \left(\frac{1+\gamma_n}{a_+}\right)^{1/k_+}$. From (4.10) we obtain $f_+^{-1}\gamma_n \sim a_+^{-1/k_+}(\ell_- b_- n)^{-1/(\ell_- k_+)}$ which yields the claimed asymptotics for $|Y_{\pm,n}|$. $\square$

Proposition 4.12. *The map $F : Y \rightarrow Y$ is a topologically mixing Gibbs-Markov map with respect to the partition $\mathcal{P}_Y$.*

Having established Lemma 4.11 the proof of Proposition 4.12 follows essentially verbatim from [CLM23] replacing the roles of $\delta_n^\pm$ with $Y_{\pm,n}$ and $\Delta_n^\pm$ with $Y_n^\pm$. For completeness we include the main steps of this argument.

Lemma 4.13. *There exists $\lambda > 1$ such that $F'(y) > \lambda$ for all $y \in Y$.*

Proof. We follow the proof of [CLM23, Proposition 3.6.]. We only consider the case $y \in Y_{+,n}$ as the case that $y \in Y_{-,n}$ is the same. Define the function $\phi := f_+^{-1} \circ f_- \circ f_+$ and notice that $\phi : Y_{+,n+1} \rightarrow Y_{+,n}$ bijectively.

Sublemma 4.14. *Let $y \in Y_{+,n}$ where n is such that $Y_{+,n} \subset U_{0+}$, then $\frac{f'(y)f'(fy)}{f'(\phi(y))} > 1$.*

Proof. If $k_+ \in (0, 1)$ then f' is decreasing on U_{0+} and so, as $y < \phi(y)$, we have that $f'(y)/f'(\phi(y)) > 1$. By construction $f(y) \in U_{-1}$ for every $y \in U_{0+}$, so $f'(fy) > 1$ and we are finished.

Assume now that $k_+ > 1$ and to ease notation set $k = k_+, a = a_+, b = b_-, \ell = \ell_-$. Setting $x = fy$ we recall that $x \in U_{-1}$ by Remark 4.6 and using the convexity of f on U_{-1} we obtain that $fx \leq f(-1) + f'(x)(x+1) = -1 + f'(x)ay^k$ which in turn implies that $\phi(y) \leq f'(x)^{1/k}y$. As $y < \phi(y)$ and as $(f'(y))/(f'(\phi(y))) = (y/\phi(y))^{k-1}$,

$$\frac{f'(y)}{f'(\phi(y))} f'(fy) \geq \left(\frac{\phi(y)}{y}\right) \left(\frac{f'(x)^{1/k}y}{y}\right)^{-k} f'(x) \geq 1.$$

□

Let $m_+ = \min\{m : Y_{+,m} \subset U_{0+}\}$. Condition F2 implies that $F'y \geq \lambda$ for all $y \in Y_{+,m}$ whenever $m \leq m_+$. The Sublemma above allows us to conclude that if $y \in Y_{+,m+1}$ and $m+1 \geq m_+$ then

$$(f^{m+1})'(y) = f'(y) \cdot f'(fy) \cdots f'(f^m y) = \frac{f'(y) \cdot f'(fy)}{f'(\phi(y))} (f^m)'(\phi(y)) \geq (f^m)'(\phi(y)),$$

and so $F'y \geq F(\phi(y))$. Proceeding inductively, for any $y \in Y_n \subset U_{0+}$, we obtain $F'(y) \geq F'(\phi^{n+1-m^+}(y)) \geq \lambda$. □

Lemma 4.15. *There exists a $C > 0$ and a $\theta \in (0, 1)$ such that*

$$\left| \log \frac{F'(x)}{F'(y)} \right| \leq C\theta^{s(x,y)}.$$

Proof. We only consider $x, y \in Y_{+,n}$ as the argument for $x, y \in Y_{-,n}$ is the same. Calculating as in (4.1) one finds

$$(4.11) \quad \left| \log \frac{(f^{n-j})'(f^j x)}{(f^{n-j})'(f^j y)} \right| \leq \left| \frac{f''u_0}{f'u_0} \right| |Y_{+,n}| + \sum_{k=1}^{n-1} \left| \frac{f''u_k}{f'u_k} \right| |X_{-,n-k}|$$

Notice the $f''(u_0)/f'(u_0) \lesssim u_0^{-1} \lesssim (f^{-1}\gamma_n)^{-1} \lesssim (1 + \gamma_n)^{-1/k_+}$ and (4.10) gives that $(1 + \gamma_n)^{1/k_+} \lesssim n^{-\alpha_-}$. So, as $|Y_{+,n}| \lesssim n^{-1-\alpha_-}$ we find that the first term in (4.11) is uniformly bounded in n . If f is C^2 on $\bar{U}_{-1}$ the second term in (4.11) is summable and whence uniformly bounded. Otherwise, by assumption F2, (4.10) and Lemma 4.11 we have that $f''(u_k)|Y_{+,n}| \lesssim (1 + \gamma_{n-k}^-)^{\ell_- - 1} |Y_{+,n-k}| \lesssim (n-k)^{-1+1/\ell_-} (n-k)^{-1-1/\ell_-} \lesssim (n-k)^{-2}$ which is summable. Thus, (4.11) is uniformly bounded and we can calculate as in (4.2) to find $\log |F'x/F'y| \leq C|Fx - Fy|$.

Thus, proceeding in the same way as in the proof of Proposition 4.2 one concludes the result with $\theta := \tilde{\lambda}^{-1}$. $\square$

Proof of Proposition 4.12. The fact that F is Gibbs-Markov with respect to $\mathcal{P}_Y$ follows from Lemmas 4.13 and 4.15. From (4.9) we obtain $F(Y_{\pm,n}) = f^n(Y_{\pm,n}) = Y$, so F is full-branched and hence topologically mixing. $\square$

Lemma 4.16. *f satisfies H3.*

Proof. We follow the argument given in Proposition 4.2. The main difference between our current setting and that of Proposition 4.2 is that $|f'|$ is no longer necessarily bounded away from 0 and $|f''|$ may not be finite at the points $\pm 1, 0 \pm$ so we have to take a little more care.

Let $K(X) = C^1(X)$, and let $\rho \in C^1(X)$. Then $Q_r^\rho(y) = \frac{\rho(x_+)}{(f^r)'(x_+)} + \frac{\rho(x_-)}{(f^r)'(x_-)}$ for $y \in Y$, where $x_\pm \in X_{\pm,r}$ is such that $f^r x_\pm = y$. Thus, $|Q_r^\rho|_\infty \leq 2|\rho|_\infty$.

Let $y, y' \in Y$ and let $x, x' \in X_\pm$ be such that $f^r x = y$ and $f^r x' = y'$. Set $X^{(r)} = \bigcup_{k=1}^r X_{-,k} \cup X_{+,k}$. Notice that (4.4) holds as before, but now

$$\begin{aligned} |(f^r)'(x) - (f^r)'(x')| &\leq |f'|_{X^{(r)}}|_\infty^{r-1} \sum_{j=0}^{r-1} |f'(f^j x) - f'(f^j x')| \\ &\leq |f'|_{X^{(r)}}|_\infty^{r-1} |f''|_{X^{(r)}}|_\infty \sum_{j=0}^{r-1} |f^j x - f^j x'|. \end{aligned}$$

Moreover, $\inf_{j=0,\dots,r-1, \zeta \in X_{-,r-j}} (f^{r-j})'(\zeta) > 0$ and so $|f^j x - f^j x'| \lesssim |f^r x - f^r x'|$, where the implied constant depends on r . Thus,

$$\left| \frac{1}{(f^r)'(x)} - \frac{1}{(f^r)'(x')} \right| \lesssim |(f^r)'(x) - (f^r)'(x')| \lesssim |y - y'|.$$

As before, $|\rho(x) - \rho(x')| \leq |\rho'|_\infty |x - x'| \leq |\rho'|_\infty |y - y'|$. Hence, by (4.4)

$$|Q_r^\rho(y) - Q_r^\rho(y')| \lesssim |y - y'|.$$

Finally, $F'(x) \geq \lambda$, so $|y - y'| \lesssim d_\theta(y, y')$ where θ is as in Lemma 4.15. Hence $Q_r^\rho \in \mathcal{B}_\theta(Y)$ as required. $\square$

Proof of Theorem 4.9. Proposition 4.12 yields H1 and Lemma 4.16 yields H3. So, it only remains to show H2 and H4. By Lemma 2.1 F preserves an absolutely continuous invariant measure μ with density h . We claim that h is continuous on Y . Given the validity of this claim we can apply the same argument as in (4.6) to conclude from Lemma 4.11 that

$$\mu(\tau^\pm = n) = \mu(Y_{\pm,n}) \sim h(0)|Y_{\pm,n}| \sim h(0)a_\pm^{-1/k_\pm}(\ell_\mp b_\mp)^{-\alpha_\pm} \alpha_\pm n^{-1-\alpha_\pm}.$$

Together with Lemma 4.10, this concludes H2 and H4.

Let $\mathcal{P}_Y^{(n)}$ denote the n^{th} refined partition of F and let $a \in \mathcal{P}_Y^{(n)}$. From the uniform expansion of F and the mean-value theorem we know that for $j \leq n$, $|F^j x - F^j y| \leq \lambda^{-(n-j)} |F^n x - F^n y|$ for every $x, y \in a$. Moreover, recalling that $\log |F'x/F'y| \leq C|Fx - Fy|$ for $x, y \in Y$ it follows that for any $x, y \in a$

$$(4.12) \quad \log \left| \frac{(F^n)'x}{(F^n)'y} \right| \leq \sum_{j=0}^{n-1} \left| \frac{(F^j)'x}{(F^j)'y} \right| \leq \frac{C}{\lambda - 1} |F^n x - F^n y|,$$

and so $|(F^n)'x/(F^n)'y - 1| \lesssim |F^n x - F^n y|$. Moreover, (4.12) implies $1/[(F^n)'y] \lesssim |a|$ for all $y \in a$. Using the standard pointwise formula for the transfer operator L_Y for F , we obtain

$$\begin{aligned} |L_Y^n 1(x) - L_Y^n 1(y)| &= \left| \sum_{\tilde{x} \in F^{-n}x, \tilde{y} \in F^{-n}y} \frac{1}{(F^n)'\tilde{x}} - \frac{1}{(F^n)'\tilde{y}} \right| \\ &\leq \sum_{\tilde{x} \in F^{-n}x, \tilde{y} \in F^{-n}y} \left| \frac{(F^n)'\tilde{y}}{(F^n)'\tilde{x}} - 1 \right| \left| \frac{1}{(F^n)'\tilde{y}} \right| \\ &\lesssim \sum_{a \in \mathcal{P}_Y^{(n)}} |x - y| |a| \lesssim |x - y|. \end{aligned}$$

Similarly, one can check that L_Y^n is uniformly bounded. The Arzelà-Ascoli Theorem then yields that $\frac{1}{n} \sum_{j=1}^{n-1} L_Y^j 1$ has a subsequence which is uniformly convergent to some Lipschitz $\tilde{h} : Y \rightarrow \mathbb{R}$. As $\tilde{h}$ is necessarily the density of an invariant measure it follows that $h1_Y = \tilde{h}$ yielding that h is Lipschitz (and hence continuous). This establishes our claim and concludes the proof. $\square$

Appendix A. Series estimates

In this appendix, we carry out some calculations required in Section 3.4.

Lemma A.1. *Let $\alpha \in (0, 1)$. Then $\lim_{n \rightarrow \infty} \sum_{j=1}^{n-1} (n-j)^{\alpha-1} j^{-\alpha} = \frac{\pi}{\sin \alpha \pi}$.*

Proof. The function $(n-x)^{\alpha-1} x^{-\alpha}$ has one critical point at $x = \alpha n$. Hence, by approximating integrals by Riemann sums on the intervals $[0, \alpha n]$ and $[\alpha n, n]$,

$$\sum_{j=1}^{n-1} (n-j)^{\alpha-1} j^{-\alpha} = \int_0^n (n-x)^{\alpha-1} x^{-\alpha} dx + O(n^{-\alpha}) + O(n^{\alpha-1}).$$

Using standard properties of Beta and Gamma functions, we obtain

$$\begin{aligned} \int_0^n (n-x)^{\alpha-1} x^{-\alpha} dx &= \int_0^1 (1-x)^{\alpha-1} x^{-\alpha} dx = B(\alpha, 1-\alpha) = \Gamma(\alpha)\Gamma(1-\alpha)/\Gamma(1) \\ &= \Gamma(\alpha)\Gamma(1-\alpha) = \frac{\pi}{\sin \alpha \pi}, \end{aligned}$$

as required. $\square$

Lemma A.2. *Let $\alpha \in (0, 1)$. Suppose that $g : \mathbb{N} \rightarrow \mathbb{R}$ satisfies $\lim_{n \rightarrow \infty} g(n) = 0$. Then*

$$\lim_{n \rightarrow \infty} \sum_{j=1}^{n-1} j^{-\alpha} g(j) (n-j)^{\alpha-1} = 0$$

and $\lim_{n \rightarrow \infty} \sum_{j=1}^{n-1} (n-j)^{\alpha-1} g(n-j) j^{-\alpha} = 0$.

Proof. Set $C = \max_n |g(n)|$. Let $\varepsilon \in (0, \frac{1}{2})$. Then

$$\begin{aligned} \sum_{j=1}^{n-1} j^{-\alpha} g(j) (n-j)^{\alpha-1} &= \sum_{1 \leq j < \varepsilon n} j^{-\alpha} g(j) (n-j)^{\alpha-1} + \sum_{\varepsilon n \leq j < n} j^{-\alpha} g(j) (n-j)^{\alpha-1} \\ &\leq C 2^{1-\alpha} n^{\alpha-1} \sum_{1 \leq j < \varepsilon n} j^{-\alpha} + (\varepsilon n)^{-\alpha} \max_{k \geq \varepsilon n} g(k) \sum_{\varepsilon n \leq j < n} (n-j)^{\alpha-1} \\ &\lesssim n^{\alpha-1} (1 + (\varepsilon n)^{1-\alpha}) + (\varepsilon n)^{-\alpha} \max_{k \geq \varepsilon n} g(k) n^{\alpha} \\ &\lesssim n^{\alpha-1} + \varepsilon^{1-\alpha} + \varepsilon^{-\alpha} \max_{k \geq \varepsilon n} g(k). \end{aligned}$$

Hence,

$$\limsup_{n \rightarrow \infty} \sum_{j=1}^{n-1} j^{-\alpha} g(j) (n-j)^{\alpha-1} \lesssim \varepsilon^{1-\alpha}.$$

The first limit follows since ε is arbitrarily small. Replacing α by $1 - \alpha$, we obtain the second limit. $\square$

Lemma A.3. *Suppose that $g(x) = o(1)$ as $x \rightarrow \infty$. Then $\sum_{j=1}^n \frac{g(j)}{j} = o(\log n)$.*

Proof. For $\varepsilon > 0$, choose $n_0 \geq 1$ such that $|g(x)| < \varepsilon$ for $x > n_0$. Then

$$\sum_{j=1}^n \frac{g(j)}{j} \leq \sum_{j=1}^{n_0} \frac{g(j)}{j} + \varepsilon \sum_{j=n_0}^n \frac{1}{j} \leq \sum_{j=1}^{n_0} \frac{g(j)}{j} + \varepsilon \log n.$$

Hence $\limsup_{n \rightarrow \infty} (\log n)^{-1} \sum_{j=1}^n \frac{g(j)}{j} \leq \varepsilon$. $\square$

Lemma A.4. *Let $g_1, g_2 : [0, \infty) \rightarrow \mathbb{R}$ with $\lim_{x \rightarrow \infty} g_i(x) = 1$ for $i = 1, 2$. Then*

$$\lim_{n \rightarrow \infty} \sum_{j=1}^{n-2} \frac{g_1(j)}{j} \frac{g_2(n-j)}{\log(n-j)} = 1.$$

Proof. Write

$$\sum_{j=1}^{n-2} \frac{g_1(j)}{j} \frac{g_2(n-j)}{\log(n-j)} = S_1 + S_2$$

where

$$S_1 = \sum_{1 \leq j \leq n/2} \frac{g_1(j)}{j} \frac{g_2(n-j)}{\log(n-j)}, \quad S_2 = \sum_{n/2 < j \leq n-2} \frac{g_1(j)}{j} \frac{g_2(n-j)}{\log(n-j)}.$$

Now,

$$\begin{aligned} |S_2| &\leq |g_1|_{\infty} |g_2|_{\infty} \frac{2}{n} \sum_{n/2 \leq j \leq n-2} \frac{1}{\log(n-j)} \\ &= |g_1|_{\infty} |g_2|_{\infty} \frac{2}{n} \sum_{2 \leq j \leq n/2} \frac{1}{\log j} \leq |g_1|_{\infty} |g_2|_{\infty} \frac{2}{n} \sum_{2 \leq j \leq n} \frac{1}{\log j}. \end{aligned}$$

But

$$\sum_{2 \leq j \leq n} \frac{1}{\log j} \lesssim \frac{n}{\log n} + \sum_{n/\log n \leq j \leq n} \frac{1}{\log j} \leq \frac{n}{\log n} + \frac{1}{\log(n/\log n)} n \lesssim \frac{n}{\log n},$$

so $|S_2| \lesssim \frac{1}{\log n}$.

By Lemma A.3, $\sum_{j=1}^n \frac{g_1(j)}{j} \sim \log n$. Hence,

$$S_1 \leq \sup_{k \geq n/2} g_2(k) \frac{1}{\log n/2} \sum_{1 \leq j \leq n/2} \frac{g_1(j)}{j} \sim \frac{1}{\log n/2} \log n/2 = 1.$$

Similarly,

$$S_1 \geq \inf_{k \geq n/2} g_2(k) \frac{1}{\log n} \sum_{1 \leq j \leq n/2} \frac{g_1(j)}{j} \sim \frac{1}{\log n} \log n/2 \sim 1.$$

Hence $S_1 + S_2 \sim S_1 \sim 1$ as required. $\square$

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